

Superior Energy Storage Performances of Polymer Nanocomposites via Modification of Filler/Polymer Interfaces

Xin Zhang,* Bao-Wen Li, Lijie Dong, Hanxin Liu, Wen Chen, Yang Shen, and Ce-Wen Nan*

Dielectric polymer nanocomposites by integration of high- E_b polymer matrix and high- $D(\epsilon_r)$ ceramic fillers have shown great potential for dielectric and energy storage applications in modern electronic and electrical systems. Interface between ceramic fillers and polymer matrix is considered as a predominant factor to determine the dielectric performances of the nanocomposites. This review analyzes the influence of the interface on dielectric responses and breakdown strength in the nanocomposites, and discusses the viability of current interface engineering strategies in improving their energy storage capabilities. Two scopes of current interface modification approaches are focused from both structural and functional considerations in the dielectric ceramics/polymer nanocomposites: first, the organic/inorganic interface compatibility can be modified to generate homogeneous dispersion of ceramic nanoparticles in polymer matrix, which is the premise of fully realizing the synergistic combination of advantages of polymer and ceramic fillers; second, regulated local electrical and dielectric behaviors in interface region enable the enhancement of dielectric properties (both high dielectric constant and high breakdown strength) in resultant nanocomposites. In the last part, some present challenges and future perspectives are proposed to utilize the interface strategy for developing high energy density ceramics/polymer nanocomposites for dielectric and energy storage applications.

systems, they have numerous applications in various fields as pulsed power supply technology,^[8,9] energy harvesting,^[10–12] inverters,^[13–15] and passive elements,^[16–18] toward both defense and civil industries. Excellent energy storage capabilities of dielectric materials including high discharged energy density and high energy efficiency have long been eagerly pursued to meet the challenges and needs of the rapid development of modern industry, i.e., the low energy density of current dielectric materials results in overburdened capacitor volume and weight in electrical power systems. The energy storage performances of dielectric materials are usually multiply determined by three major electrical and dielectric parameters, which are categorized as:

- 1) The breakdown strength E_b ;
- 2) The relative permittivity (also called dielectric constant) ϵ_r or electric displacement D (the electric displacement is related to the polarization P by $D = P + \epsilon_0 E$, here $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$ is the vacuum permittivity);
- 3) The dielectric loss $\tan \delta$.

1. Introduction

Dielectric materials with superior electrical, mechanical, and thermal properties are the main enabler of electrostatic energy storage devices, known as dielectric capacitors, and have garnered substantial both academic and commercial interests due to their typical capability of delivering an ultrahigh power density, low loss, and withstanding high operating voltage.^[1–7] Being playing critical roles in modern electronic and electrical

In general, as illustrated in **Figure 1**, the discharged energy density of dielectric materials can be determined as

$$U_e = \int_{D_{\max}}^0 E \, dD \quad (1)$$

where E is the applied electric field, and the energy efficiency is calculated by

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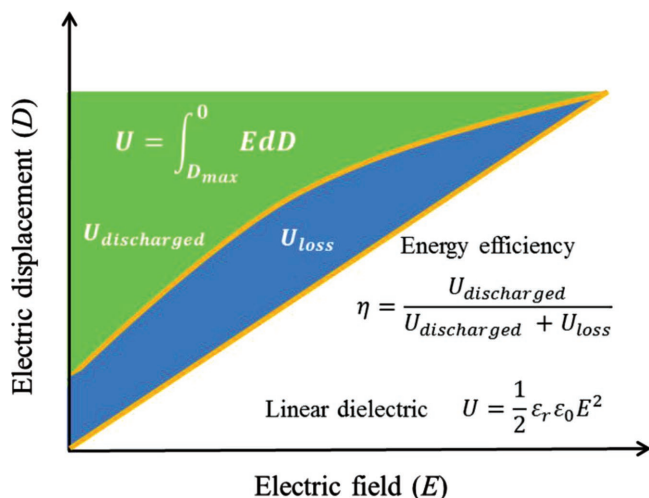


Figure 1. Schematic illustration of discharged energy density (green shadowed area) and energy loss (blue shadowed area) from the electric displacement (D)–electric field (E) loop. For linear dielectrics, the discharged energy density is determined from the triangular area (green + blue).

$$\eta = \frac{U_{\text{discharged}}}{U_{\text{discharged}} + U_{\text{loss}}} \quad (2)$$

For linear dielectrics, whose dielectric constant is invariable to the electric field, and the energy density can be formulated as $U_e = 1/2 \epsilon_r \epsilon_0 E_b^2$, according to which, the enhancement of either ϵ_r or E_b can lead to improvement of energy density. The dielectric constant is an intrinsic parameter of dielectric materials associated with the types and amounts of polarizabilities, which indicates the ability and extent of polarization. Usually, four types of polarizations exist in general dielectric materials including electronic, ionic, dipolar, and interfacial (space-charge) polarizations. Breakdown strength is the critical value of electric field which makes dielectric materials turning from insulating state to conducting state. The breakdown of dielectrics under electric field is a complicated process synthetically affected by electrical, mechanical, and thermal issues. Given its quadratic relationship with energy density, high breakdown strength is crucial to achieve high energy density. However, the real dilemma is that dielectric constant and breakdown strength for monophase dielectric materials are mutually restricted, which can be described with an empirical relationship as $E_b \sim \epsilon_r^{-0.5}$.^[19] Such paradox between E_b and ϵ_r is unavoidable for any monophase polymeric or ceramic dielectric materials because both the parameters are related to the intrinsic bonding conditions, which severely hinders the improvement of energy density.

Dielectric ceramics are one major category of conventional dielectric materials. Ceramics with perovskite structure such as BaTiO_3 , usually possess dielectric constant as high as hundreds or even thousands but low breakdown strength, and they also suffer some other adverseness such as inflexibility, high mass density, and complicated processing conditions associated with high temperature sintering,^[20–22] limiting their applications for high energy density capacitors. Compared to ceramics, the counterpart polymers are preferable candidates for high energy density dielectric capacitors due to their superiority in ultrahigh breakdown strength ($>500 \text{ MV m}^{-1}$), low mass density, flexibility,



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and easy fabrication process with low cost and high throughput. However, most polymers are suffering from intrinsic low dielectric constant,^[23] making them still fall short of achieving an enough high U_e . To this end, polymer nanocomposites by integration of high- E_b polymer matrix and high- $D(\epsilon_r)$ ceramic fillers have shown great potential in achieving superior energy storage performances by concomitant enhancement of both $D(\epsilon_r)$ and E_b .

Even in the face of apparently combining the advantageous properties of both polymer matrix and ceramic fillers, the ultimate energy storage properties of the resultant nanocomposites are also determined by a number of other factors including filler features (e.g., concentration, size, shape, and spatial distribution), possible changes in the polymer matrix due to fillers, and polymer/filler interfaces. Among them, interfaces have been considered as predominant features to determine the dielectric behaviors of nanocomposites, especially when the size of filling particles is reduced to nanoscale.^[24–27] The interface effects on the dielectric properties of nanocomposites have been extensively studied and can be described from two aspects. On the one hand, the interfacial structural characteristics such as inorganic/organic compatibility, polymer/filler interactions, polymer chain configuration, and particle surface states can determine the final dispersion or aggregation of ceramic nanoparticles in the polymer matrix. On the other hand, the abnormal local

interfacial electrical and dielectric characteristics, e.g., interfacial electronic states, their roles acting as traps or scattering points, charge density, redistribution of charges and electric fields, enable to regulate the dielectric and breakdown behaviors of nanocomposites beyond the bulk properties of polymer and ceramic fillers. In polymer nanocomposites filled with nano-sized fillers, those unique interfacial dielectric characteristics are amplified by the significant proportion of interfaces.

This review aims to analyze possible influence of interface on dielectric responses and breakdown strength in ceramics/polymer nanocomposites for high energy storage applications, and discusses the viability of interface modification in improving their energy storage performances. We summarize current approaches employed to modify the interfaces between polymer and ceramic fillers, which achieve not only improved dispersion of nanofillers in polymer matrix, but also the accessible enhancement of dielectric and energy storage performances in resultant nanocomposites from regulated local interfacial dielectric and breakdown behaviors.

2. High Energy Density Ceramics/Polymer Nanocomposites

2.1. Dielectric Polymers

Serving as the matrix in the dielectric nanocomposites, the dielectric and energy storage capability of polymers themselves determine the comprehensive performances of dielectric composites to a large extent, polymers with high breakdown strength, high dielectric constant, and low dielectric loss are highly desired.^[28] Both linear and nonlinear dielectric polymers have been extensively employed to prepare polymer nanocomposite dielectrics for electrical energy storage. Typically, linear dielectric polymers have intrinsic low dielectric loss and high insulativity, and have long been considered as industrial candidate materials for applications in thin film capacitors. However, due to the low dipole moment and low dipole density in the molecular chains, most linear dielectric polymers are suffering from low dielectric constant hence small electrical displacement, which severely limits the achievable energy density. The most commercially available dielectric polymer used in

film capacitors is the biaxially oriented polypropylene (BOPP) film. PP is a nonpolar polymer with an ultrahigh breakdown strength over 700 MV m^{-1} and low dielectric loss less than 0.02%. In spite of these, the energy density of the BOPP films is still rather limited to $1\text{--}2 \text{ J cm}^{-3}$, owing to its low dielectric constant ($\epsilon_r = 2.2$).^[29–31] It is feasible to improve the dielectric constant while still maintaining low dielectric loss in linear polymer by introducing high polar groups into the molecular chains,^[32] and typical examples of polar linear polymers are polyurea and polythiourea. Aromatic polyurea (ArPU) is a typical strong polar linear polymer with a large dipole moment of 4.5 Debye.^[33] A relatively high dielectric constant of 4.2 and high breakdown strength of 800 MV m^{-1} were achieved in aromatic polyurea thin films fabricated by vapor deposition polymerization. The polymer film exhibits linear dielectric responses even under high electric fields and high temperatures, hence the highest energy density can reach over 12 J cm^{-3} with a charge–discharge efficiency above 90%.^[34] Aromatic polythiourea (ArPTU) possesses even higher dipole moment of 4.89 Debye compared with ArPU, leading to a dielectric constant of 4.5, which is higher than most linear dielectric polymers. The dipole coupling is quite weak in ArPTU because of the amorphous phase structure, thus there is no polarization hysteresis even at high electric field of 1.1 GV m^{-1} . As a consequence, a much higher energy density of 22 J cm^{-3} could be achieved in ArPTU thin film at the breakdown electric field.^[33,35,36]

Nonlinear polymers with polar groups in their molecular chains usually exhibit higher dielectric constants compared to the linear ones. Poly(vinylidene fluoride) (PVDF)-based polymers are one family of the most frequently used dielectric materials for high energy density capacitors. The fluorine atoms in the molecular chains have large electronegativity and small size, which offer both high dipole moment and high packing density of C–F bonds,^[37] as a result, the dielectric constant of PVDF can reach around 10. The dielectric and electrical properties of PVDF are largely dependent on the polymer chain configuration. PVDF is a semicrystalline polymer with multiple crystalline phases, which usually can be divided into four polymorphs (see Figure 2a), including α (in *trans*–*gauche* conformation TGTG), β (in all-*trans* planar zigzag conformation TTTT), γ (in a conformation of three *trans* linked to a *gauche* TTTG), and δ (a polar version of α phase).^[38] PVDF of β phase exhibits the

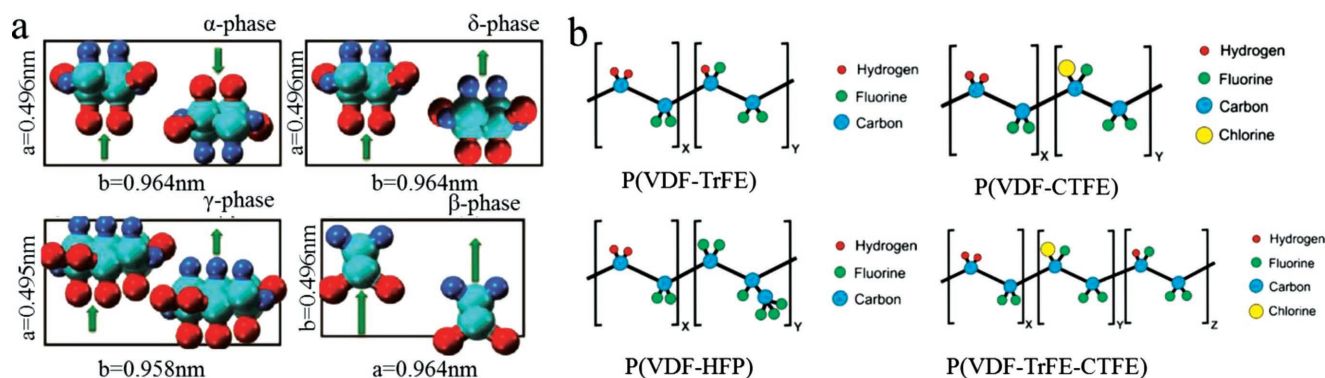


Figure 2. a) Unit cells of PVDF crystals with four different phases viewed along *c*-axis. Reproduced with permission.^[38] Copyright 2012, American Chemical Society. b) Schematic illustration of polymer chain structures of P(VDF-TrFE), P(VDF-CTFE), P(VDF-HFP), and P(VDF-TrFE-CTFE). Reproduced with permission.^[46] Copyright 2014, Elsevier.

strongest polarity as well as ferroelectric hysteresis, while PVDF of α or γ phases usually leads to less polarized phases as well as weaker hysteresis.^[39] In the PVDF of β phase, the CF_2 groups are aligned in the same direction because of the all-*trans* chain configuration, leading to the highest chain dipole moment and producing the highest polarization with electric field applied. Among the various phases, only β phase exhibits ferroelectricity with thermodynamic stable spontaneous polarization. The β phase of PVDF can be obtained by uniaxially stretching, melt crystallization under high cooling rates, and crystallization from solvent like dimethylformamide.^[40–42] In spite of the highest polarization, it is still difficult to achieve high discharged energy density in PVDF of β phase, because the strong ferroelectric hysteresis leads to high remnant polarization and low discharged efficiency. Different from β phase, the α phase is a nonpolar phase where the CF_2 groups are packed in opposite directions, thus the dipole moments are canceled by each other, exhibiting zero net dipole moment. By poling the α phase at room temperature, the δ phase can be obtained. Among all the phases, γ phase, with a stronger polarity than α phase but weaker than β phase, has shown the most promising dielectric performances for energy storage applications. A common and effective approach to obtain PVDF of γ phase is quenching the materials in ice water from high temperature over 200 °C.^[43,44] The existence of γ phase enables PVDF withstanding a high breakdown strength, and exhibiting a paraelectric like *D–E* loops with relatively low remnant polarization and high discharge energy efficiency. The discharged energy density obtained in γ -PVDF can reach as high as 14 J cm^{−3}.^[45]

In order to further enhance the dielectric properties of PVDF, monomers including trifluoroethylene (TrFE), chlorotrifluoroethylene (CTFE), chlorofluoroethylene (CFE), and hexafluoropropylene (HFP) have been introduced into the VDF chains to form PVDF-based copolymers and terpolymers, which exhibit excellent dielectric and electrical properties.^[23,46] Poly(vinylidene fluoride-trifluoroethylene) (P(VDF-TrFE)) is a copolymer of PVDF with strong ferroelectricity.^[47] The TrFE incorporated into VDF chains acts as defects that help to enhance the crystallinity and stabilize the β phase. Upon high energy radiation, the ferroelectric nature of P(VDF-TrFE) can be tuned and transferred into relaxor ferroelectrics,^[48–53] for the electron irradiation can break up the coherent polarization domain into nanopolar regions.^[54] For instance, with increasing the strength of irradiation, the ferroelectric domain sizes reduce from 15 to 5 nm, and gradually become unobservable, resulting in a relaxor or even a nonpolar phase in polymer with dramatically reduced hysteresis.^[55] Another approach to reduce the hysteresis and acquire relaxor ferroelectricity in polymer is the chemical modification of the molecular chain structure of ferroelectric polymers. By introducing CTFE, CFE monomers into P(VDF-TrFE) polymer chains, relaxor ferroelectric properties can also be achieved in terpolymer poly(vinylidene fluoride-trifluoroethylene-chlorotrifluoroethylene) (P(VDF-TrFE-CTFE))^[56] and poly(vinylidene fluoride-trifluoroethylene-chlorofluoroethylene) (P(VDF-TrFE-CFE)).^[57–60] Similar to the irradiation effects mentioned above, the CTFE and CFE monomers act as defects to effectively break up the large ferroelectric domains into nanodomains, and consequently lead to relaxor ferroelectricity with suppressed remnant

polarization.^[38] However, the maximum discharged energy density of relaxor ferroelectric polymer is still limited due to the early saturation behaviors, specifically, the electrical displacement tends to be saturated at elevated electric field, and shows no further obvious increment with continuously increasing the applied electric field. For instance, P(VDF-TrFE-CFE) terpolymer has an ultrahigh dielectric constant over 50 at room temperature, while the discharged energy density is only 10 J cm^{−3}.^[58] Compared with ferroelectric and relaxor ferroelectric polymers, CTFE and HFP monomer-modified PVDF copolymers have weaker ferroelectricity and hysteresis, which are more in favor for achieving high energy density. For instance, poly(vinylidene fluoride-chlorotrifluoroethylene) (P(VDF-CTFE)) copolymer holds a lower dielectric constant of 13 and also lower hysteresis, which delivers a high energy density of 25 J cm^{−3} at electric field of 600 MV m^{−1}.^[61] The remnant polarization and hysteresis can be further suppressed in the polystyrene (PS)-modified P(VDF-CTFE)-*g*-PS graft copolymer, where the PS side chains are segregated to the periphery of PVDF crystals and serve as confining interfacial layer.^[62] Another copolymer poly(vinylidene fluoride-hexafluoropropylene) (P(VDF-HFP)) exhibits paraelectric like *D–E* loops with suppressed hysteresis and early saturation, the incorporation of HFP monomers can also largely enhance the mechanical properties of polymer,^[63] resulting in a much elevated breakdown strength over 700 MV m^{−1}, the highest energy density achieved in P(VDF-HFP) can reach over 25 J cm^{−3}.^[64] However, all these nonlinear polymers usually exhibit higher dielectric loss than the PP.

2.2. Ceramic Nanofillers

The nanofillers employed for the reinforcement of dielectric polymer nanocomposites can be mainly divided into two categories: nonconducting fillers and conducting fillers. When using conducting nanoparticles as fillers, the dielectric and electrical performances of nanocomposites are primarily determined by the percolation behavior. When the concentration of conducting fillers reaches critical volume fraction, the particles tend to contact with each other to form conducting paths in the nanocomposites, which is associated with a transition from insulating to conducting state of the dielectric nanocomposites. At a concentration very near but still below the percolation threshold, a dramatically increased dielectric constant can be achieved in such nanocomposites. Using this approach, various nano- or micro-sized conducting fillers, such as metal nanoparticles,^[65–72] carbon materials (graphene, carbon nanotubes, carbon black),^[73–84] and conductive polymers,^[85,86] have been employed to achieve a high dielectric constant in composite materials.

In such nanocomposites, dielectric constant can be effectively enhanced by precisely controlling the concentration of nanofillers near percolation threshold. For example, the dielectric constant of Ni/PVDF composite could be dramatically increased to 400 as volume fraction of Ni particles increased to 17%.^[87] And usually, the percolation threshold can be decreased when the nanoparticles have large aspect ratios, such as nanowires, nanofibers, and nanosheets. In PVDF-based nanocomposites filled with carbon nanotubes (CNTs), a dielectric constant

over 1000 was achieved with only 8% volume fraction of CNT.^[88] However, dielectric constant will drop down over the percolation threshold because the nanocomposites transform from insulating to conducting state. Efforts have been made to ease the substantial change of electrical properties near the percolation. Shen et al.^[89] modified interfacial condition between the conducting filler and insulating matrix. An amorphous carbon layer was coated onto the surface of Ag nanoparticles before embedding Ag nanoparticles into the epoxy matrix. The existence of C layer could impede the direct intercontact between Ag nanoparticles, thus the high dielectric constant could be maintained even when the concentration of Ag exceeding the percolation threshold. In spite of the much enhanced dielectric constant, it is still quite difficult to achieve high energy density in nanocomposites filled with conducting nanoparticles. The dilemma encountered in the percolative nanocomposites is that such enhancement is always associated with a critical transition from insulator to conductor, resulting in much high dielectric loss and leakage current. Moreover, the significant difference of conductivity between filler and polymer matrix induces strong distortions of electric field near the interface region, resulting in very limited breakdown strength. Considering the very limited achievable energy density, the nanocomposites with conducting fillers are not to be discussed in detail in this review.

Compared with conducting nanoparticles, nonconducting nanoparticles have been used widely for these composite approaches. In such nanocomposites, there is no substantial change of electrical properties with the appearance of percolation, which provides more tolerance and possibilities to tune the dielectric properties and achieve a high energy density. In recent years, various nonconducting nanoparticles with different dimensions have been introduced into polymer matrix. These nanoparticles mainly include high- ϵ_r ceramic nanoparticles such as BaTiO₃,^[90–104] TiO₂,^[105–112] ZrO₂,^[113–116] (Ba_xSr_{1-x})TiO₃,^[43,117–121] CaCu₃Ti₄O₁₂,^[122,123] Pb(Zr_xTi_{1-x})O₃,^[124–126] naturally nanosized silicate minerals (attapulgite, montmorillonite, clay),^[127–134] and other materials with excellent electrical and thermal properties, such as boron nitride.^[1,135–143]

For these nanofillers, a number of factors including the intrinsic dielectric and electrical properties, geometry shapes, and dimensions, concentrations, sizes, would affect the final dielectric and energy storage performances of polymer nanocomposites. For the purpose of compensating the low dielectric constant of polymer matrix, high- ϵ_r ceramic nanoparticles are one of the most widely used nanofillers. Ferroelectric ceramics with perovskite structure possess intrinsic high dielectric constant up to hundreds and even thousands owing to the spontaneous polarization. Usually, with reducing the size of perovskite ceramic particles, there would be a phase transition from tetragonal to cubic, thus decreasing the ferroelectricity as well as dielectric constant, then nanosized ceramic particles have relatively lower dielectric constant compared with bulk ceramics, however, it is still much higher value than that of polymers. The incorporation of ceramic nanoparticles can effectively lead to an enhanced dielectric constant compared with pure polymer matrix. While the high- ϵ_r ceramics also bring about some adverse effects, one of the most concerned issues is the decline of breakdown strength, especially when the concentration is high. The contact between nanoparticles, structural

imperfections, and electric field distortions at the interfaces all lead to increased dielectric loss and decreased breakdown strength. For example, by employing BaTiO₃ nanoparticles as dielectric reinforcement in P(VDF-HFP) matrix, the dielectric constant was increased from 12 for pure polymer to near 30 with 40% volume fraction of BaTiO₃ nanoparticles in the resultant nanocomposite, while the breakdown strength showed a sharp drop at loading volume fraction between 10% and 20%.^[92]

Using 1D nanofibers with large aspect ratio as fillers is in favor of an enhanced dielectric breakdown strength within certain volume fraction. Nanofibers can lead to increased path tortuosity in the electrical treeing process during breakdown, when they are along the in-plane directions perpendicular to the external electric field.^[94,96,144,145] Various nanofibers synthesized by either electrospinning or hydrothermal processes such as BaTiO₃,^[94,146,147] TiO₂,^[44,108,148] Pb(Zr_xTi_{1-x})O₃,^[125,126] and (Ba_{1-x}Sr_x)TiO₃,^[43,118] have been introduced into the polymers. These in-plane-oriented nanofibers enable enhancement of breakdown strength compared to pure polymer within certain volume fraction. Moreover, nanofibers also show even more efficiency in enhancing the dielectric constant of nanocomposites than nanospheres because of a larger achievable dipole displacement.^[149] Similar enhancement of breakdown strength could also be achieved by using 2D nanosheets as nanofillers, such as layered montmorillonite,^[134] boron nitride (BN),^[135,136] TiNaNbO₃.^[150] For example, in P(VDF-TrFE-CFE)-based nanocomposites reinforced with hexagonal boron nitride nanosheets (BNNS),^[135] the uniformly dispersed BNNS resulted in greatly improved breakdown strength up to 610 MV m⁻¹ with the BNNS filler content of 12%, which is a 70% increment compared to the pure terpolymer (362 MV m⁻¹). The enhanced E_b could be first ascribed to excellent mechanical strength of nanocomposites, where the BNNS served as robust scaffold to hamper the electromechanical failure. Second, the high insulating network of BNNS could reduce the leakage current and space-charge conduction. Moreover, the remnant displacement was also reduced by BNNS, suppressing the conduction loss. As a result, high discharged energy density over 20 J cm⁻³ as well as high discharge efficiency of 80% was achieved in the nanocomposites at an electric field of 600 MV m⁻¹.

2.3. Preparation of Nanocomposites

The most conventional and simple method to prepare polymer nanocomposites is direct mixing of polymer and nanofillers, including melt mixing and solution mixing. Melt mixing is usually associated with a high temperature melting process, nanoparticles are filled into melting polymer and dispersed by shear forces, and the nanocomposites are obtained by subsequent pressing or extrusion molding process.^[130] This method is mainly suitable for thermoplastic polymers, which can be pliable and moldable above softening and melting temperatures. The drawback of such a mixing process is that the mobility of polymer chains is still limited even at the molten state. The counterpart solution mixing is a liquid phase based method by dispersing the nanoparticles in polymer-dissolved solution, which provides the level of molecular mixing.^[151] The nanocomposites can be molded into films from solution mixtures

by solution-casting process. In such a solution process, surface modifications are usually needed to enhance the incompatibility between inorganic nanoparticles and organic polymer matrix, thus avoiding the aggregation of nanoparticles. The surfaces of nanoparticles are naturally hydrophilic while the polymers are hydrophobic, and the liquid media offers proper environment to modify the polymer/nanoparticle interfaces by tuning the surface potentials and surface polarities. Another common approach to fabricate polymer nanocomposites is in situ polymerization of monomers in the presence of nanofillers. In a typical process, the nanofillers are immersed in a liquid monomer or monomer solution, and the monomers are initiated and polymerized into polymer by heat, radiation, or a proper initiator between those nanofillers.^[8] One benefit of such process is the formation of strong interaction between polymer and fillers when the initiating points are sited on the filler surfaces.

Beyond polymer matrix and nanofillers, interfaces are the third dimension of nanocomposite approach, which plays an even more significant role in determining the properties of nanocomposites. The physics and chemistry of polymer/filler interface will be elaborated in detail to present a comprehensive understanding of interface effects in polymer nanocomposites in the next section.

3. Physics and Chemistry of Interfaces in Dielectric Nanocomposites

3.1. Interface Characteristics in Dielectric Nanocomposites

In general, the interfacial region in polymer nanocomposites can be characterized to extend from the surface of the nanoparticle,

through the modification layer (if present) and interfacial polymer layers with modified chain structures, until to the host matrix polymer. Thus, the interfacial region can be considered as a shell around the particle with a certain thickness. The content of interfacial region becomes significant and even predominant in the nanocomposites when the size of nanofillers is reduced to nanoscale. The volume fraction of interfaces with a finite thickness can be quantitatively analyzed in the 0–3 type nanocomposites filled with spherical nanoparticles (**Figure 3a**). The volume fraction of the interfaces ($f_{\text{interface}}$) is related to the diameter (d) of nanoparticle, interface thickness (t), and particle volume fraction (f_{particle}) with the following expression

$$f_{\text{interface}} = f_{\text{particle}} \left[\left(1 + \frac{2t}{d} \right)^3 - 1 \right] \quad (3)$$

According to it, with fixed interface thickness, decreasing the diameter of nanoparticles will lead to significant increase of the volume fraction of the interfaces. For instance, as shown in **Figure 3b**, when f_{particle} is fixed at 5% and t is fixed at 5 nm, the incorporation of nanoparticles with $d = 100$ nm leads to less than 10% of interface volume fraction, while the $f_{\text{interface}}$ dramatically increases to over 50% for nanoparticles with $d < 10$ nm.

Another important feature of interfaces in nanocomposites is that the properties (electrical, dielectric, thermal, and mechanical) at interface regions differ from either the polymer matrix or the nanofillers. The interface region can thus be defined as the range over which the electrical and dielectric properties change from the value of ceramic nanoparticles to that of polymer matrix. In general, three categories of possible behaviors may occur across the polymer/nanoparticle interface. As illustrated

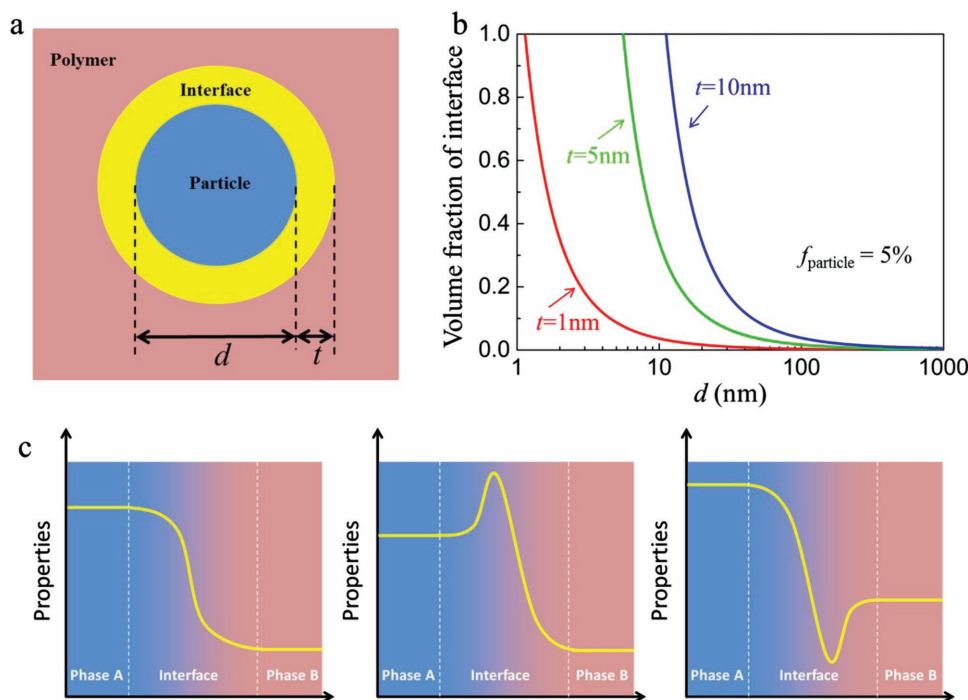


Figure 3. a) Schematic illustration of the filler–polymer interface structure in nanocomposite consisting of spherical nanoparticle. b) Calculated variation of interface volume fraction with the diameter of nanoparticles possessing different interface thicknesses, the volume fraction of nanoparticles is fixed at 5%. c) Possible behaviors of interfacial properties ranging across the interfaces.

in Figure 3c, considering the nanoparticle and matrix as two uniform material phases A and B, then conventionally, the properties of interface phase are limited within the two phases and gradually change from a value of phase A to phase B. While in other cases, the properties at interface exhibit abnormal higher or lower values than that of both phases. Considering the significant volume fraction of interface region, the unique properties at interface would be amplified and even become predominant to determine the properties of nanocomposites. The effects of interfacial characteristics in dielectric ceramics/polymer nanocomposites, e.g., nanoparticle surface states, polymer chain configurations, inorganic/organic compatibility, local interfacial electrical and dielectric behaviors, on the dielectric and energy storage properties of dielectric nanocomposites have been extensively studied, and a few conclusions can be proposed:

- 1) Interfacial incompatibility exists due to the hydrophilicity of ceramic filler surface and hydrophobicity of polymer, resulting in aggregation of nanoparticles.^[151,152]
- 2) The interfaces induce the change of polymer molecular structures initiating from the interfacial region to the matrix (free volume fraction, mobility, crystallinity, and configuration of polymer chains), which affects the dielectric properties of matrix polymer as well as the nanocomposites.^[153–156]
- 3) The modified electronic states of interfaces create traps or change the depth of traps, the trapping of charges further influences the local conductivity, space-charge polarization, and breakdown behaviors.^[157–164]
- 4) The interfaces serve as scattering points, which increase the breakdown strength of nanocomposites.^[165,166]
- 5) The difference in dielectric constant and electrical conductivity between ceramic filler and polymer creates mismatch at the interfaces, leading to local redistribution of charges and electric fields, and also creating Maxwell–Wagner interfacial polarization.^[167–169]
- 6) The mobile charges in the diffuse layer (in terms of double layer model) lead to local conductivity being associated with percolation behaviors when the overlap of interface region occurs.^[25,159,170]

3.2. Interface Models

In order to describe the interfacial electrical and dielectric behaviors, Lewis^[24,25] extended the diffuse electrical double layer model into the polymer nanocomposites. After the introduction of nanoparticles into the polymer matrix, the surface of nanoparticles becomes charged due to the different Fermi levels or chemical potentials between polymer and nanoparticles. The charged surface induces screening ionic charges in surrounding polymer, which in turn suppresses the charge accumulation on particles. In general, the screening ionic charges are generated in two ways on the polymer side. On one hand, they could be induced by the polarization of polymer involving both the electronic polarization and orientation of permanent dipoles, which usually occurs when polymer has polar components. On the other hand, the migration and redistribution of mobile charges on the polymer side due to Coulomb attraction could lead to the formation of so-called electrical double layer. **Figure 4a,b** shows the distribution of charges across the nanoparticle/polymer

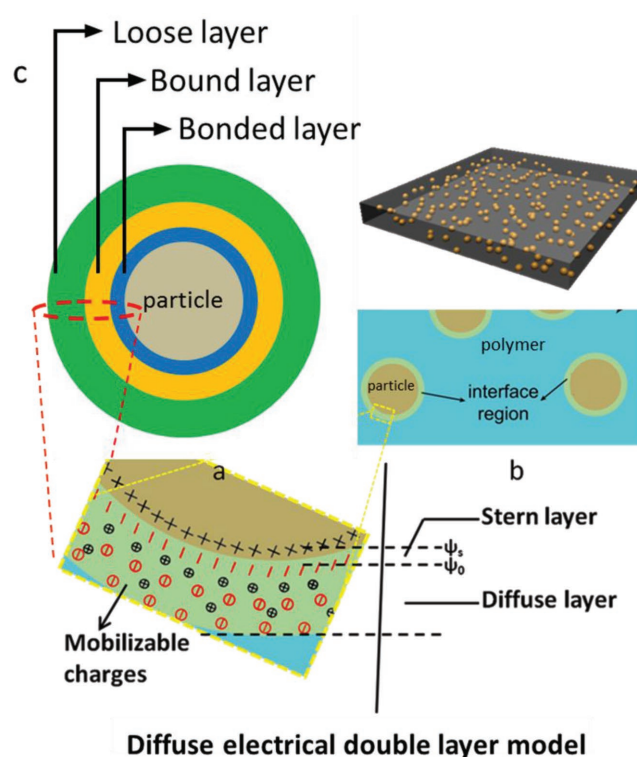


Figure 4. a,b) The diffuse electrical double layer model in polymer nanocomposites filled with positively charged particles. c) Schematic illustration of multicore model for nanoparticle/polymer interfaces.

interface when the nanoparticle is positively charged inside the nanocomposite. On the particle side, a layer exhibiting positive potential ψ_s is formed by the surface states related with immobile charged impurities, trapped carriers, mobile electrons, and holes in the nanoparticle. Next to this layer, the negatively charged counterions or dipoles are tightly absorbed to the particle surface to form the so-called Stern or Helmholtz double layer, and outer this layer is Helmholtz plane (OHP) with electrical potential of ψ_0 . Further extending the OHP into polymer matrix is the Gouy–Chapman diffused layer, which is formed outer and around the Stern layer and distributed by both negative and positive ions. The extent of this layer is decided by the ion concentration in polymer as well as the potential difference between the particle surface and the polymer matrix. The Gouy–Chapman layer filling with mobile charges plays a dominant role in affecting the dispersion of nanoparticles and determining the dielectric and electrical properties of nanocomposites. The distribution of charges in this layer is related to the electrical potential $\psi(r)$ across the interface region and varies with the distance from the particle surface, which can be described with a combined Poisson–Boltzmann equation

$$\nabla^2 \psi(r) = -e\epsilon^{-1} \sum_i z_i n_i(\infty) e^{-z_i e \psi(r)/kT} \quad (4)$$

where ϵ is the effective permittivity of the layer, z_i and $n_i(\infty)$ are the ion valency and concentration of ion species i in polymer matrix, k is the Boltzmann constant.

To further describe the chemical and physical formation of the interfaces, especially to build the linkage from chemical

or physical structures to electronic structures, Tanaka et al.^[26] proposed a multilayered core model for a spherical particle embedded in polymer matrix (Figure 4c). In this model, three layers across the interface region are distinguished and specified as bonded layer (the first layer), bounded layer (the second layer), and loose layer (the third layer), and all three layers are just overlapping with the Gouy–Chapman diffuse layer from the electrical double layer model. The first layer is considered as a transition layer and tightly bonded to both inorganic particles and polymer matrix by coupling agent. The second layer is constructed by polymer chains that strongly interacted with the first layer and nanoparticles, and the thickness of this layer is decided by the strength of polymer–particle interaction and usually in the range of 2–9 nm. In the third layer, the chain conformation, mobility, crystallinity, and free volume of polymer are different from the bulk matrix due to the loose coupling and interaction to the second layer.

3.3. Characterization of Local Electrical and Dielectric Behaviors of Ceramics/Polymer Interfaces

It is of great significance to explore the local electrical and dielectric properties of interfacial regions in the ceramics/polymer nanocomposites, which is the key to understand how the interfaces influence the dielectric properties of nanocomposites. One of the very important factors to determine the local electrical properties is the interface serving as traps, and the trapping of charges leads to regulated local charge mobility and conductivity, which is quite related to interfacial (space-charge) polarization and breakdown behaviors. In order to explore the origin of traps at interfaces, Kubyshkina et al.^[158,171] calculated the electronic structures of ceramic nanoparticle surface and interface region in modeled system of polyethylene/MgO nanocomposites using density functional theory (DFT). The electronic structures calculated for bulk MgO and alkane, hydroxylated MgO surface, silanol-terminated MgO surface, interfacial region between MgO and alkane chains are shown in Figure 5. As seen from Figure 5b, the hydroxylated surface possesses the smallest bandgap, and the further Si treatment results in a wider bandgap than the untreated hydroxylated surface. However, there would still be hydroxylated surface parts present on the Si-treated MgO surface because the Si treatment is never ideal, and such surface states in the interface region could be localized in space and act as traps for charges. After the surface modification of MgO nanoparticles, the contact between polymer and particles does not produce new electronic states, while the polymer band bending could lead to excessive electrons near the interface region and these electrons are captured in the low energy states or traps with restricted mobility.^[171] The formation of traps at interfaces reduces charge mobility, and the localized traps also induce counteracting electric field, thus leading to higher barrier for charge injection from electrodes at the electrode/insulator interfaces, and consequently contributing to increased insulativity and improved dielectric breakdown behaviors. Besides theoretical calculations, the direct observation of interfacial electrical behaviors has also been conducted by Borgani et al.^[172] They investigated the effects of traps at interfaces on the local interfacial charging and discharging

behaviors in the low-density polyethylene (LDPE)–based nanocomposites with Al₂O₃ nanoparticles as fillers. With the help of intermodulation electrostatic force microscopy (ImEFM) (Figure 5c,d), mapping of the single nanoparticle surface potential at nanometer scale under different DC biases was feasible. As shown in Figure 5e, owing to the shallow traps induced by the modified electronic states at the polymer/nanoparticle interfaces, the nanoparticles would be positively charged with the migration of holes from tip to the particle surface, thus potential difference existed across the interface even with no bias applied. Then, a positive DC bias enabled the injection of holes from the tip to the traps more energetic, thus more traps were filled and the surface potential of nanoparticle was further increased. In contrast, the holes could be extracted from the tip when negative DC bias was applied, resulting in empty traps and no contrast of potential was still remaining between the particle and polymer matrix. This work identified how the traps influence the charge injection and extraction behaviors as well as the electrical properties of the nanocomposites.

The local dielectric constant of ceramics/polymer nanoregion can also be calculated with DFT. For example, Shi and Ramprasad^[173] modeled the local and position-dependent of dielectric response at SiO₂/polymer interface in the nanocomposites of α -quartz phase SiO₂ and alkane chain (C₁₂H₂₅) as the polymer. The calculated dielectric constants (static and optical) of SiO₂ and polymer in the interior region agreed well with the bulk values of single component. However, the dielectric constant at the interface (mainly on SiO₂ side) was obviously enhanced, owing to the altered oxidation states of Si and C atoms near the interface. Yu et al.^[169] also calculated the dielectric properties at the interfaces in isotactic polypropylene (iso-PP)–based nanocomposites filled with Al₂O₃ and PbTiO₃ nanoparticles (Figure 6a,b). Models of both single bare Al₂O₃ and PbTiO₃ slabs, and their interfaces with iso-PP were constructed and calculated. Across the Al₂O₃/iso-PP interface, it has demonstrated that both the static and optic dielectric constants in the interior of the Al₂O₃ slab and iso-PP were similar to the bulk values. Meanwhile, the dielectric constant was enhanced in the subsurface region of Al₂O₃ side, but decreased on the iso-PP side, and such a transition region across the interfaces was only restricted within 6–7 atomic layers. At the PbTiO₃/iso-PP interface, the dielectric constant on the PbTiO₃ side was also enhanced at the subsurface but suppressed when approaching the outmost surface. Compared with Al₂O₃, this ferroelectric PbTiO₃ surface was more affected by the attachment of iso-PP chains because its dielectric constant was greatly dependent with the electrostatic boundary conditions. In particular, the bonding at the polymer/ferroelectric surface would affect the surface charges and hence the depolarization field for the ferroelectric PbTiO₃. The calculated results indicated that at the typical untreated polymer/nanoparticle interface, the dielectric constant was mainly determined by the particle side and usually improved dielectric constant could be achieved in the subsurface region of nanoparticle owing to modified electronic states and more polarizable subsurface layer. While, the polymer side was less significant and the dielectric constant was usually suppressed due to the increased nanovoids and free volume induced by low packing density of polymer chains.

The reduced dielectric constant on the polymer side can also be characterized with a direct detection of the local interface

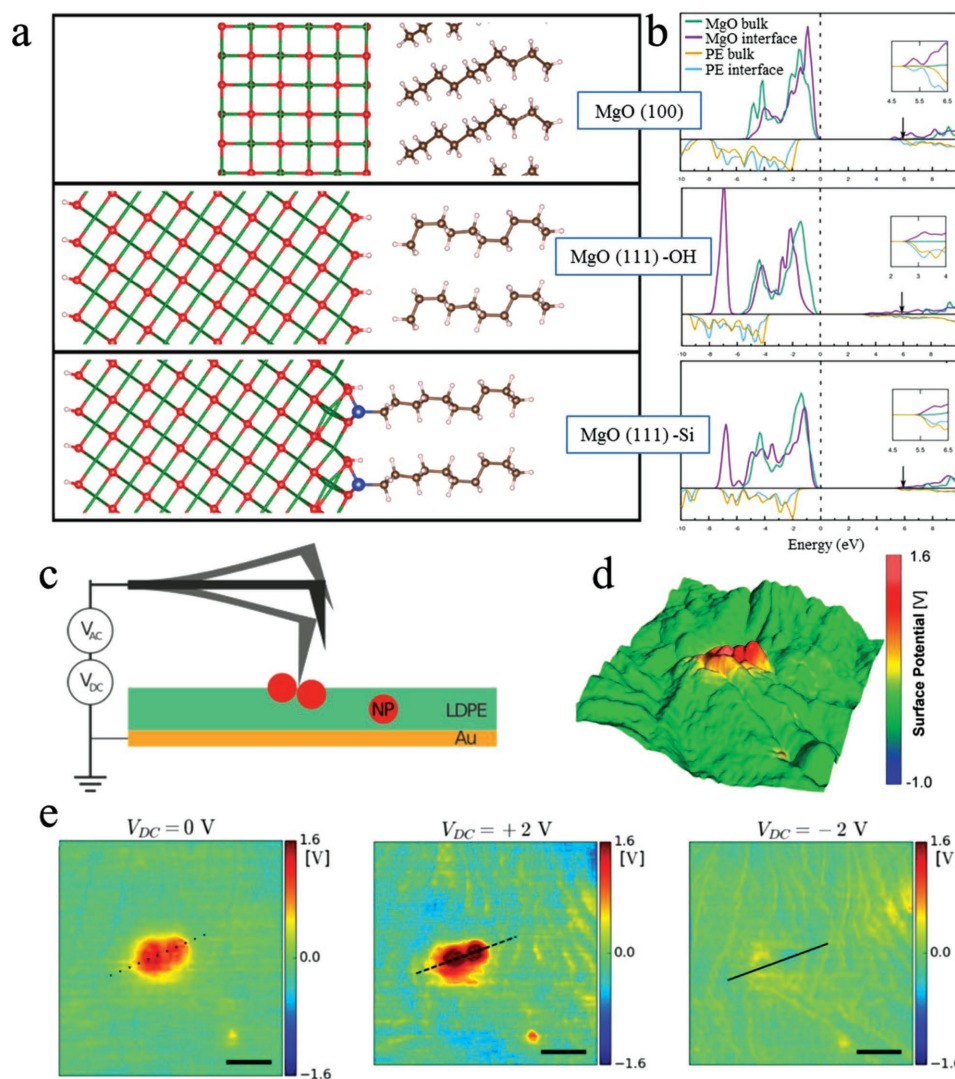


Figure 5. a) Geometric illustration of modeled MgO/polyethylene systems including MgO(100)–polyethylene interface, hydroxylated MgO(111)–polyethylene interface and Si-treated MgO(111)–polyethylene interface (magnesium atoms are shown in green, oxygen in red, hydrogen in white, carbon in brown, silicon in blue colors). b) The corresponding projected densities of states of the interfaces, zero energy is corresponding to the valence band of bulk MgO (6.6 eV), with respect to the vacuum level. The inset figures are the magnified conduction band edges and the arrows points to the bottom of the conduction band of bulk MgO. a,b) Reproduced with permission.^[171] Copyright 2017, American Institute of Physics. c) Schematic illustration of the ImEFM experiment process. d) 3D mapping of the surface potential of LDPE/Al₂O₃ nanocomposites. e) Surface potential of Al₂O₃ nanoparticles and nearby regions with zero, positive, and negative DC bias applied. c–e) Reproduced with permission.^[172] Copyright 2017, American Chemical Society.

dielectric response in polymer nanocomposites. He and co-workers^[174] conducted the nanometer scale investigation of interface between LDPE and TiO₂ with electrostatic force microscopy (EFM). As shown in Figure 6c, the signals detected from the nanocomposites could be categorized into several cases including: “matrix,” “bump,” and “exposed bump,” which corresponded to different structures and components. “Bump” was suggested as the polymer layer of interface regions, “matrix” and “exposed bump” represented the LDPE and TiO₂ nanoparticle, respectively. Based on the EFM mapping signals, the average relative dielectric constants of these cases were calculated as 2.250, 1.944, 2.546 for “matrix,” “bump,” and “exposed bump,” respectively. That is to say, the dielectric constant of polymer at interfacial region was decreased compared with the matrix polymer. The interface effects on decreasing the local

dielectric constant were further identified with finite element method. It was found that the simulation results disagreed with the experimental results without considering the interface. After introducing an interface layer with low dielectric constant outside the nanoparticle in the model, the calculated electrostatic force of “bump” was the smallest compared to other cases, which agreed well with the experimental results (Figure 6g). Thus, both the experimental and simulated results validated that the local dielectric constant was reduced on polymer side at the interfacial regions, which was mainly due to the reduced mobility of polymer chains and the low packing density.

As seen, the simulated and characterized results of ceramics/polymer interfaces exhibit abnormal and distinguished local structures and properties (electrical and dielectric) compared with polymer matrix and ceramic fillers, which have direct

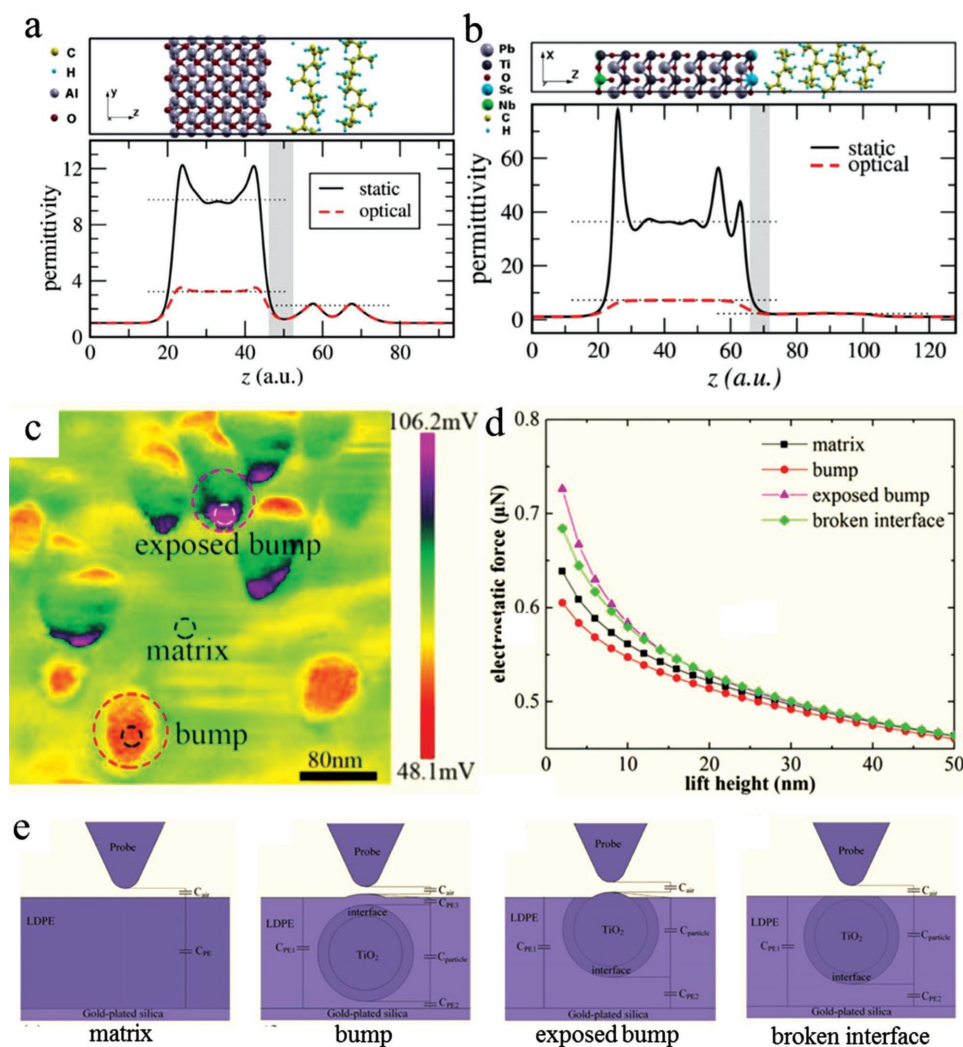


Figure 6. a) The supercell of polypropylene/AlO₂ interface model and corresponding calculated optical and static dielectric permittivity profiles. b) The supercell of polypropylene/PbTiO₃ interface model and corresponding calculated optical and static dielectric permittivity profiles. a,b) Reproduced with permission.^[169] Copyright 2009, American Physical Society. c) EFM mapping signals of different cases from the LDPE/TiO₂ nanocomposites. d) Calculated electrostatic forces of four different cases. e) Capacitance structures of four cases as “matrix,” “bump,” “exposed bump,” and “broken interface.” c–e) Reproduced with permission.^[174] Copyright 2016, Nature.

bearing on the ultimate properties of nanocomposites. It is thus feasible to improve the dielectric and energy storage properties of nanocomposites by interface modifications. Current interface modification approaches are mainly focused on improving dispersion of ceramic nanoparticles in polymer matrix, and also enhancing the local interfacial dielectric properties. These effective and viable modification approaches will be classified and elaborated in the next two sections.

4. Interface Modification for Improved Interfacial Compatibility

4.1. Difficulties of Handling the Nanoparticles

Nanoparticles with large surface energy are extremely easy to aggregate in the manufacturing process of polymer

nanocomposites, thus deteriorating their performances. The interface incompatibility also leads to inferior interaction between fillers and polymer, which induces structure imperfections such as voids and pores inside the nanocomposites, resulting in high dielectric loss and low breakdown strength. Thus, it is important to modify the filler/polymer interface with the goal of realizing good interfacial compatibility, which is essential to achieve the desired electrical and dielectric performances of nanocomposites. For nanocomposites fabricated by solution-based method, manipulating the stability of nanoparticles in liquid solutions is decisive to obtain the homogeneous dispersion of nanoparticles in resultant nanocomposites. The dispersion phenomena in aqueous media can be described by the derjaguin–landau–verwey–overbeek (DLVO) model,^[175,176] i.e., the interactions or forces (V_T) between nanoparticles are multiply determined by van der Waals attractive force (V_{WA}) and repulsive electrostatic force (V_{ER}) from the electrical double layer. The relationship

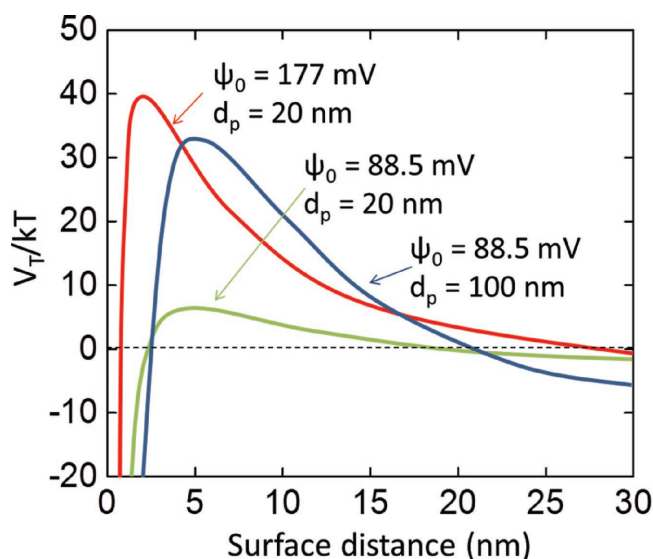


Figure 7. Potential curves calculated by DLVO theory. Reproduced with permission.^[152] Copyright 2009, Hosokawa Powder Technology Foundation.

can be expressed as $V_T = V_{ER} + V_{WA}$. When nanoparticles are of electric neutrality, they tend to aggregate by the attractive van der Waals force ($V_{WA} < 0$). When the particles are charged and surrounded by diffuse electrical double layer, repulsive electrostatic force ($V_{ER} > 0$) is generated by the overlapping of the interfacial double layer region. Only if the comprehensive force exhibits repulsive ($V_T > 0$) by the coordinating and balancing of the two forces, the particles can be stabled in the liquid system. It is feasible to control the dispersion of particles by tuning the surface potential ψ_0 and counterion concentration in the electrical double layer, while it becomes much difficult in the case of nanosized particles. **Figure 7** shows the relationship between the particle size and surface potential calculated by DLVO theory,^[152] where the total potential has been normalized by kT (k is Boltzmann constant and T is the absolute temperature) to indicate the potential barrier for the aggregation of nanoparticles. When the surface potential is fixed at 88.5 mV, the value of $V_T/kT > 30$ is obtained with the particle size of 100 nm, which is large enough to prevent the aggregation of nanoparticles. While when the particle size reduces to 20 nm, the barrier decreases to $V_T/kT < 10$, which makes the particles easier to aggregate, and a much higher surface potential up to 177 mV is needed to obtain a potential barrier $V_T/kT > 30$ for particles of 20 nm in size.

Except for V_{WA} and V_{ER} , another important factor to influence the aggregation of nanoparticles is that the surface of inorganic fillers is naturally hydrophilic, while the organic polymer is hydrophobic. Thus, repulsive force by hydrophilic interaction ($V_{HR} > 0$) and attractive force by hydrophobic interaction ($V_{HA} < 0$) also contribute to the final V_T , which has been considered in extended DLVO theory.^[177] It is necessary to modify the surface of nanoparticles to enhance the organic/inorganic compatibility, and also generate a repulsive force between nanoparticles by tuning the surface potential in the liquid solution during the fabrication process. And, the solution environment enables various modification approaches such as surface hydroxylation, chemical treatment with surface modifier, polymeric dispersant or grafting of polymer. These approaches

are further discussed in detail in the following sections. While for nanocomposites fabricated through melting process, the dispersion of nanoparticles is largely determined by the shear forces applied, which is not within the scope of our discussion.

4.2. Surface Hydroxylation

Surface hydroxylation has been one of the simplest surface modification techniques to improve the dispersion of ceramic nanoparticles inside the polymer matrix.^[117,119,178–181] For most high- ϵ_r ceramic nanoparticles such as BaTiO₃, the surface is lack of reactive functional groups, it is thus difficult to induce strong interactions (chemical bonding) between the nanoparticles and polymer. Therefore, most of the surface modifiers and polymers can only be physically absorbed on the surface by electrostatic attraction of van der Waals force.^[180] Hydroxylation generates hydroxyl (–OH) groups on the particles, and these –OH groups can chemically bond with surface modifiers or even directly interact with polymer chains.

For example, as illustrated in **Figure 8a**, BaTiO₃ nanoparticles can be treated in aqueous solution of H₂O₂ to introduce hydroxyl groups on the BaTiO₃ surface.^[179] When the hydroxylated particles were mixed with PVDF polymer, the F atoms on the polymer chains would interact with the –OH groups on the particle surface by hydrogen bond. The strong interaction between hydroxylated BaTiO₃ nanoparticles and polymer chains makes the nanocomposites more uniform with few pores compared with nanocomposites filled with crude BaTiO₃ nanoparticles (Figure 8b,c). Moreover, the hydroxyl groups on the particle surface could also lead to suppressed leakage current, dielectric loss, and increased breakdown strength of polymer nanocomposites.^[179,181] Except for the substantially enhanced interface compatibility and nanoparticle dispersity, the –OH groups on surface enable the nanoparticle to act as nucleating agent,^[119] hence crystallinity of polymer matrix could be improved. For instance, after incorporation of hydroxylated Ba_{0.6}Sr_{0.4}TiO₃ nanofibers into PVDF matrix, the crystallization temperature has been increased from 133 °C for pure polymer to 136 °C for nanocomposites filled with 10 vol% of nanofibers. Instead of directly interacting with the polymer matrix, the –OH groups also help to form strong interaction between the surface modifier with nanoparticles. For example, the BaTiO₃ nanoparticles were prehydroxylated before coating with hydantoin epoxy resin layer onto the nanoparticles.^[182]

Instead of direct hydroxylation of particle surface by using H₂O₂, Wang and co-workers^[183] reported an alternative to generate –OH groups onto the particle surface by the formation of Ba–OH. Specifically, TiO₂ nanoparticles were refluxed in aqueous CO-free barium hydroxide solution under Ar atmosphere, which resulted in the formation of Ba–OH on the TiO₂ surface. The dispersion of TiO₂ nanoparticles has been greatly improved in nanocomposites with minimized agglomeration due to the presence of Ba–OH on the surface.

4.3. Surface Modifiers

Chemical surface modifiers, e.g., saline coupling agent, dopamine,^[184,185] ethylenediamine, ammonium sulfate, phosphonic

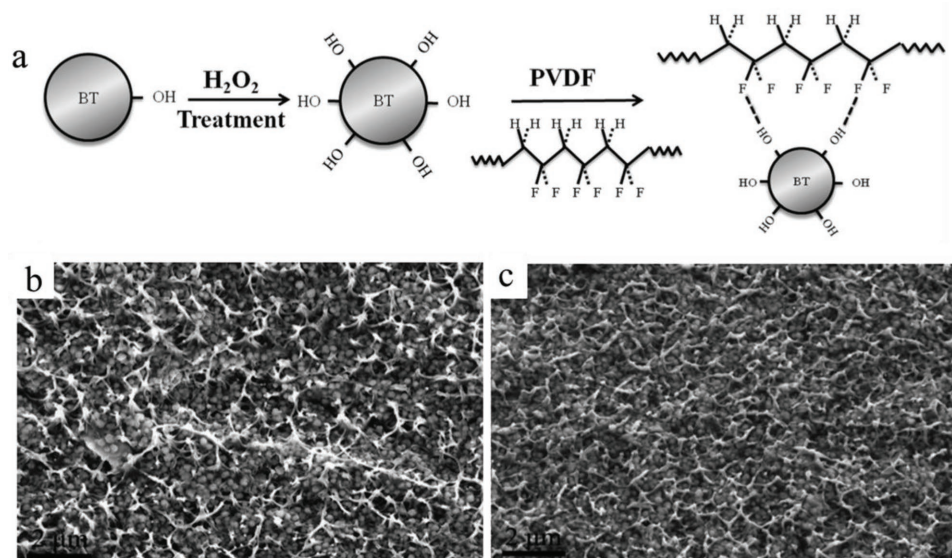


Figure 8. a) Schematic illustration of the hydroxylation process of BaTiO₃ nanoparticles. Scanning electron microscope (SEM) images of fractured cross-surface of PVDF nanocomposites filled with b) crude BaTiO₃ nanoparticles and c) hydroxylated BaTiO₃ nanoparticles. Reproduced with permission.^[179] Copyright 2011, American Chemical Society.

acid, have been extensively studied and employed to modify the polymer/filler interface in nanocomposites. The typical molecular structure of surface modifier is illustrated in **Figure 9**, which usually consists of three parts, a group to interact with particles (R_{particle}), a group to interact with polymer chains (R_{polymer}), and a middle connecting part (R). The R_{particle} and R_{polymer} usually have different polarities of hydrophilicity or hydrophobicity. The principle of choosing a modifier is to satisfy concurrently strong interaction with both nanoparticles and polymers by different functional groups at two sides.

Song et al.^[94] modified the surface of BaTiO₃ nanoparticles and BaTiO₃ nanofibers with dopamine (simplified as Dopa@BTNPs and Dopa@BTNFs), and further studied the effects of surface modification on the dielectric constant and breakdown strength of polymer nanocomposites. A 5 nm thick layer of dopamine was formed on the surface of BaTiO₃, which was characterized and confirmed by transmission electron microscope (TEM) images (**Figure 10a,d**). Such a surface dopamine layer resulted in homogeneous dispersion of both BaTiO₃ nanoparticles and nanofibers in the epoxy matrix, making the nanocomposites uniform and of high quality, while aggregation of nanoparticles was observed in the composites filled with unmodified fillers (**Figure 10b,c,e,f**). The dopamine-modified BaTiO₃ nanoparticles and nanofibers were more effective in increasing the dielectric constant of nanocomposites at all

volume fractions compared with unmodified ones, and the dielectric loss remained less than 0.04 (**Figure 10g**). Moreover, the dopamine modification could further increase the breakdown strength of nanocomposites. For example, the highest breakdown strength was achieved in nanocomposite filled with 2 vol% of Dopa@BTNFs (**Figure 10f**). Such enhancement of breakdown strength was attributed to the strong interaction between polymer chains and fillers. The polymer chains were tightly bonded to the surface-modified BaTiO₃, the mobility of the polymer chains was thus decreased and the transfer of mobile charges through the loose polymer chains was suppressed. In Jiang and co-workers' work,^[144] dopamine was employed for the surface modification of four different ceramic nanofibers including Na₂Ti₃O₇, TiO₂, BaTiO₃, and SrTiO₃, and all the modified nanofillers exhibited homogeneous dispersion in the PVDF matrix. So the effects from intrinsic properties of different nanowires could be investigated without interferences from structure imperfection in the nanocomposites. Among them, Dopa@BaTiO₃ was demonstrated to be the best candidate for potential dielectric and energy storage applications.

Another frequently used modifier with amine groups is ethylenediamine, which has been extensively employed to modify the interfaces in PVDF-based nanocomposites for dielectric energy storage.^[96,97,100] For example, Tang and Sodano^[43] investigated the energy storage properties of P(VDF-HFP)-based

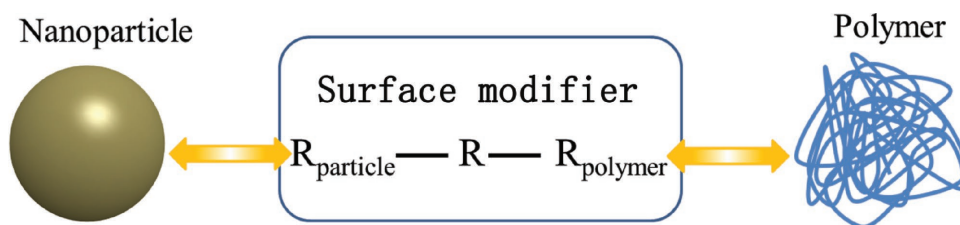


Figure 9. Schematic illustration of structure of surface modifier.

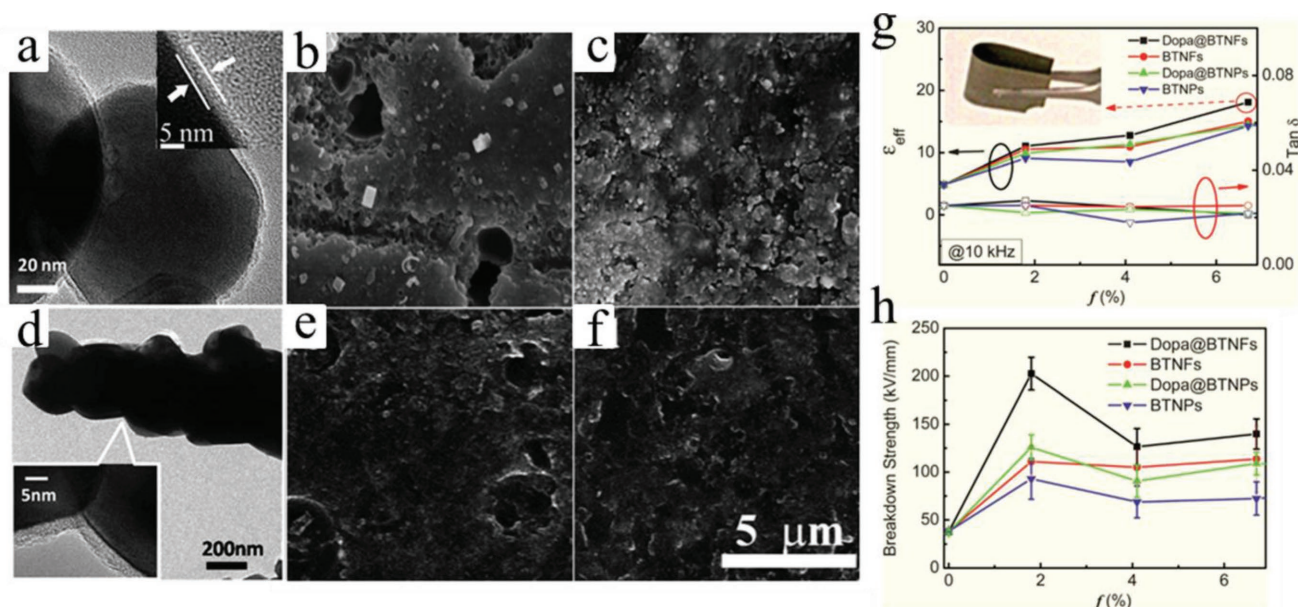


Figure 10. The TEM images of a) BaTiO₃ nanoparticles and d) BaTiO₃ nanofibers coated with dopamine layer. The surface SEM images for epoxy-based nanocomposites filled with b) BaTiO₃ nanoparticles, c) Dopa@BaTiO₃ nanoparticles, e) BaTiO₃ nanofibers, and f) Dopa@BaTiO₃ nanofibers at volume fraction of 6.7 vol%. g) Variations of dielectric constants and dielectric loss with the volume fraction of different BaTiO₃ fillers. h) Variations of breakdown strength with the volume fraction of different BaTiO₃ fillers. Reproduced with permission.^[94] Copyright 2012, Royal Society of Chemistry.

nanocomposites by using Ba_{0.2}Sr_{0.8}TiO₃ (BST) nanowires as nanofillers and ethylenediamine as the surface modifier (Figure 11a). The formation of amine groups on the BST nanofibers strengthened their interaction with PVDF, resulting in improved dispersion and charge uniformity at the nanofiber/polymer interface. The nanocomposites filled with homogeneously dispersed BST nanowires exhibited a high energy density of 14.86 J cm⁻³ at 450 MV m⁻¹ (Figure 11b).

Silane coupling agent is another most accessed modifiers for the interface between inorganic oxide nanoparticles and polymers, due to their capability of building the linkage between the polymer and inorganic fillers.^[186–193] In a typical process of silane coupling agent interacting with the nanoparticle surface, the alkoxide groups (Si–OCH₃) are hydrolyzed by water and the Si–OH groups are thus generated, then the silane coupling

agents can be introduced to the particles by the condensation reaction between the Si–OH groups and –OH groups from the particle surface.^[186] Meanwhile, the functional groups on the other side of molecular chains can interact with various matrix polymers. Thus, strong interaction between polymer and nanoparticles is built. For example, Dang et al.^[187] modified the surface of BaTiO₃ nanoparticles with 3-aminopropyltriethoxysilane (KH550) in PVDF-based nanocomposites. KH550 is a commonly used silane coupling agent with the –Si(OC₂H₅)₃ group to interact with BaTiO₃ via condensation of –OH, and meanwhile, the –NH₂ group can interact with the PVDF through the formation of F–H bond (as illustrated in Figure 12a). The nanocomposite filled with BaTiO₃ modified by 1 wt% KH550 exhibited favorable uniformity and homogeneous dispersion of nanoparticles, while the nanocomposite

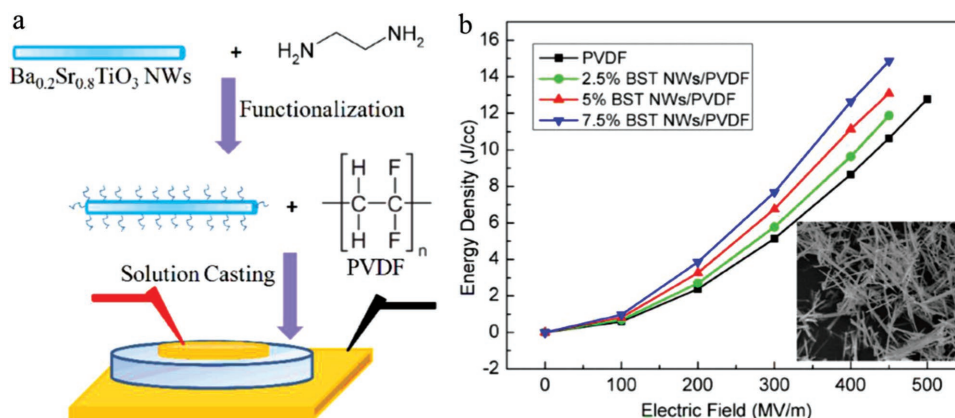


Figure 11. a) Schematic illustration of surface functionalization of nanowire, nanocomposite preparation, and characterization of nanocomposite film. b) Energy density of the nanocomposites with different filler volume fractions as a function of the electric field, inset is the SEM image of surface-modified nanowires. Reproduced with permission.^[43] Copyright 2013, American Chemical Society.

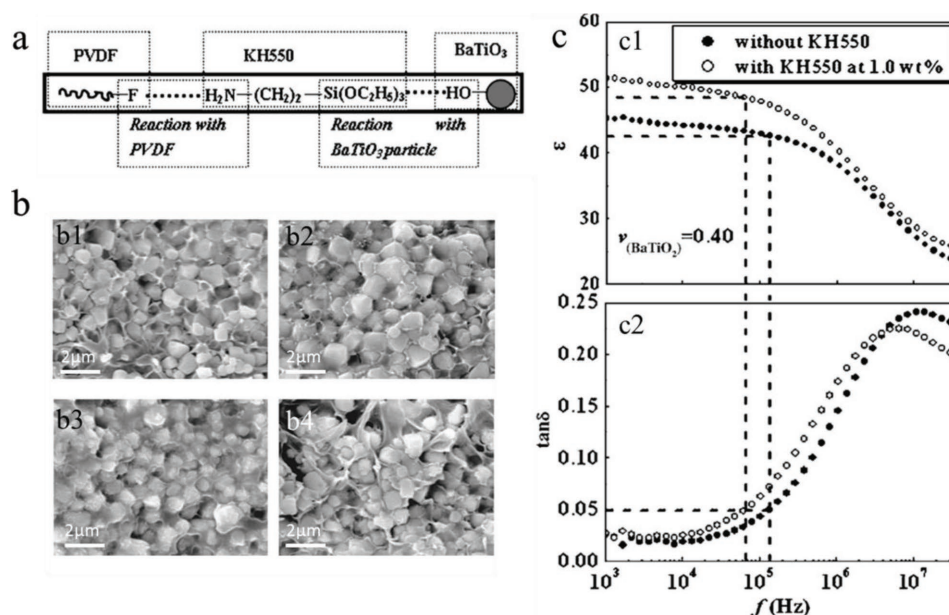


Figure 12. a) Schematic image of chemical reaction process of KH550 with both surfaces of BaTiO₃ and PVDF. b) SEM images of fractured surfaces of BaTiO₃/PVDF composites (40.0 vol% of BaTiO₃), in which BaTiO₃ was modified with different concentrations of KH550: b1) 0 wt%, b2) 1 wt%, b3) 2 wt%, b4) 5 wt%. c) Frequency dependence of c1) the dielectric constant and c2) the dielectric loss of the BaTiO₃/PVDF composites (40.0 vol% of BaTiO₃), in which BaTiO₃ was treated without (solid circle) KH550 and with (hollow circle) KH550 at 1.0 wt%. Reproduced with permission.^[187] Copyright 2006, American Institute of Physics.

showed poor quality with aggregation and pores when BaTiO₃ nanoparticles were unmodified or modified with more than 1 wt% KH550 (Figure 12b). Therefore, the content of KH550 must be controlled for an expectable morphology of the nanocomposites. Finally, as shown in Figure 12c, the dielectric constant of the PVDF nanocomposite filled with 1.0 wt% KH550-treated BaTiO₃ was improved by 20%, in comparison with the nanocomposite without KH550 treatment. However, the detriment is that the presence of KH550 also resulted in increased dielectric loss. Silane coupling agent has also been employed for modifying the surface of BN.^[188] BN has attracted substantial interests due to its superior thermal conductivity, high insulativity, and 2D geometric advantage of large aspect ratio, which make BN a preferred candidate as filler of polymer nanocomposites to achieve regulated thermal behaviors and improved breakdown strength. By modifying the surface of BN platelets with silane coupling agent KH550, the surface hydrophilicity was thus decreased, leading to lower in-plane orientation and improved interfacial adhesion with the matrix polymer.^[188]

When silane coupling agent was used as the modifier, solvent such as water or alcohol is necessary to provide —OH group for the hydrolysis process of alkoxide groups to generate the —Si—OH groups. Thus, the type and environment of solvents can largely affect the reaction process of coupling agent with inorganic particles. For example, it has been proved that the pH of water has a large effect on the surface structure of nanoparticles, hence affecting the surface interactions with silane coupling agent. When modifying the surface of silica nanoparticles with a silane coupling agent (3-glycidoxypolytrimethoxysilane, GPTMS), the optimum pH value for the reaction was confirmed as 3, which could lead to the optimum polymerized structure of GPTMS on the particle surface.^[190]

Moreover, the modification efficiency of silane coupling agent also largely depends on the type of solvent. The chemisorbed amounts of GPTMS on the particle surface in three different solvents including water, ethanol, and xylene were quantitatively analyzed. Water was found to be the optimum choice, enabling the largest reacted amount of GPTMS, which provided the highest hydrolysis rate of silane alkoxides compared with ethanol or xylene.^[186] In spite of the advantages in interfacial enhancement by the strong interaction between polymer and nanoparticles, it should be noted that the concentration of silane coupling agent needs to be properly controlled to achieve excellent dielectric performances in nanocomposites, since an excessive concentration not only creates structure imperfections but also increases the loss tangent of composites.^[108,187]

When using molecular surfactant as surface modifier in polymer nanocomposites, one of the drawbacks is that residual free molecules can lead to high leakage current and dielectric loss, especially when the modifier is ionic. Thus, robust chemical bonds from the modifier to the particle surface are highly desired. Perry and co-workers^[93] identified that phosphonic acid can form robust, functional organic shells on the surface of BaTiO₃ nanoparticles. Two phosphonic acids including {2-[2-(2-methoxyethoxy)ethoxy]ethyl}phosphonic acid (PEGPA) and pentafluorobenzyl phosphonic acid (PFBPA) (Figure 13), were employed to modify the interfaces between BaTiO₃ nanoparticles with polycarbonate (PC) and P(VDF-HFP) host materials. PEGPA has a tri(ethylene glycol) chain to compatibilize BaTiO₃ nanoparticles with polar, hydrophilic resins such as PC. And, the fluorinated aryl group of PFBPA enabled the compatibility with fluoroelestomers and common organic solvents, which suits for the P(VDF-HFP) matrix. The uniformity of nanocomposites filled with surface-modified

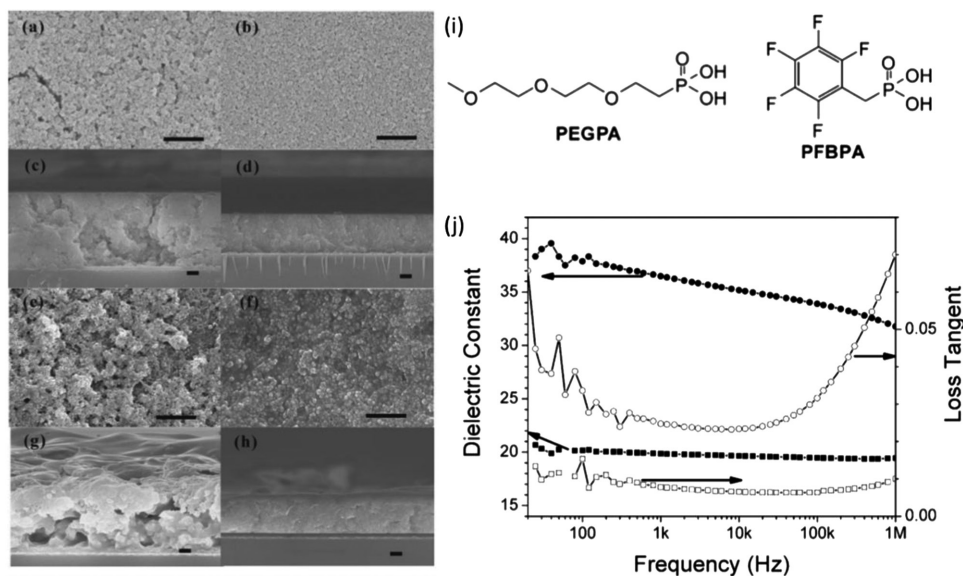


Figure 13. SEM images of surface and cross-section morphologies of nanocomposite films. a,c) BaTiO₃/PC, b,d) PEGPA-BaTiO₃/PC, e,g) BaTiO₃/P(VDF-HFP), and f,h) PFBPA-BaTiO₃/P(VDF-HFP). All the scale bars are 1 μ m. i) Structures of two phosphonic acid: PEGPA and PFBPA. j) Frequency dependence of dielectric permittivity and loss tangent of PEGPA-BaTiO₃/PC (squares) and PFBPA-BaTiO₃/P(VDF-HFP) (circles) nanocomposites. Reproduced with permission.^[93]

BaTiO₃ nanoparticles has been greatly enhanced compared to the unmodified BaTiO₃ particle-filled nanocomposites, which exhibited structure imperfections such as aggregation, cracks, and pinhole defects (Figure 13). Finally, a large dielectric constant of ≈ 37 , low dielectric loss of <0.07 were achieved in P(VDF-HFP)/PFBPA-BaTiO₃ nanocomposite (Figure 13j). The improved dielectric properties were contributed by interface enhancement from two aspects: first, the phosphonic acids could suppress the aggregation of nanoparticles, thus reducing the percolating pathways; second, the phosphonic acids could minimize the concentration and mobility of charge carriers in the interface region. However, even in the case of homogeneous dispersion of nanoparticles, excessive concentration of nanofillers would still result in deteriorated dielectric properties, e.g., reduced dielectric constant and increased dielectric loss, owing to an insufficient amount of polymer to fill the interstitial free volume generated by the random packing of the particles.^[92]

4.4. Grafting of Polymer

Besides chemical modification with surface modifiers via various long- or short-distance interactions, grafting polymer to the surface of nanoparticles has also been widely accepted, which is usually associated with a polymerization process. Two methods categorized as “grafting to” and “grafting from” have been reported to graft polymer chains onto the surface of nanoparticles.^[6] Both the methods usually result in core-shell-structured nanoparticle/polymer nanocomposites. The polymer shell can not only provide strong interaction with both nanoparticles and polymer matrix, thus enhancing the uniformity, but also serve as the matrix directly. Such approach is in particular feasible for the fabrication of polymer nanocomposites with high loading of nanoparticles, and the core-shell structure enables good

dispersion of nanofillers, resulting in high dielectric constant, low dielectric loss, and high breakdown strength.^[6]

In the “grafting to” method, the as-prepared polymer chains directly react with the particle surface by the interaction between end groups from polymer chains and functional groups from the particle surface. For example, in the process of preparing BaTiO₃@PVDF-HFP-poly(glycidyl methacrylate) (GMA) nanocomposites by “grafting to” route.^[194] The PVDF-HFP-GMA polymer was prepared by modifying the P(VDF-HFP) chains with GMA, and the BaTiO₃ nanoparticles were functionalized with amino-terminated silane molecules. Then, the as-prepared polymer was introduced onto the surface of BaTiO₃ by covalent bonding between epoxy groups from GMA and amino groups from silane molecules. The core-shell-structured BaTiO₃@PVDF-HFP-GMA nanocomposites exhibited homogeneous dispersion of nanoparticles and enhanced dielectric properties. Jiang and co-workers^[195] also reported a “grafting to” strategy by combination of reversible addition-fragmentation chain transfer (RAFT) polymerization and thiol-ene click reaction. Thiol-terminated polymer chains with three different molecular weights were prepared by RAFT polymerization, and then introduced to the vinyl-functionalized BaTiO₃ nanoparticles by thiol-ene click reaction. These core-shell-structured polymer@BT nanocomposites showed significantly enhanced dielectric constant while relatively low level of dielectric loss compared with pure polymer. This work also demonstrated the molecular weight of polymer chains and grafting density of core-shell-structured nanocomposites are quite related to the dielectric response and energy storage properties. High molecular weight of polymer chains usually leads to low grafting density and high dielectric loss.

Also by “grafting to” method, Jung et al.^[90] grafted polystyrene-*block*-poly(styrene-*co*-vinylbenzylchloride) (PS-*b*-PSVBC) copolymer on the surface of BaTiO₃ nanoparticles to form a double-shell structure, these core-shell-structured particles

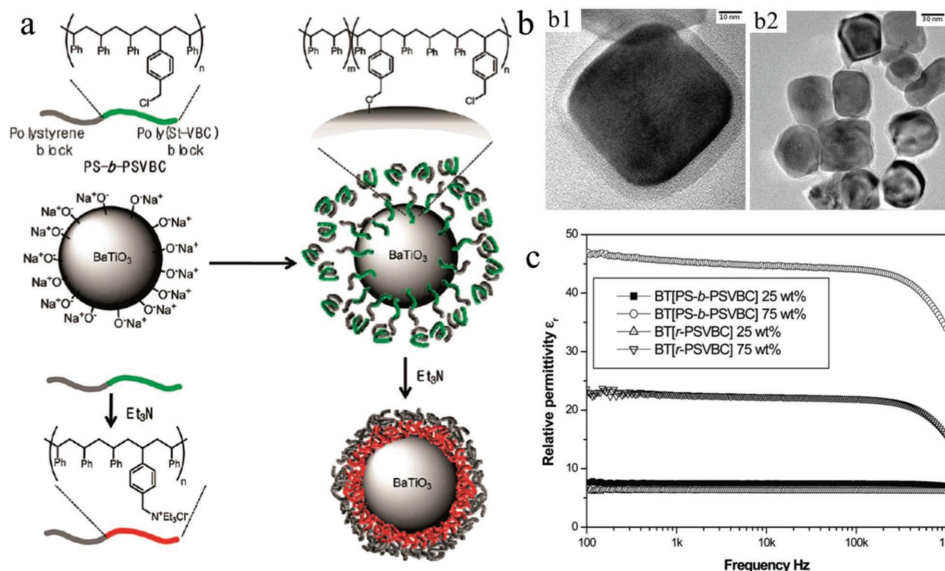


Figure 14. a) Schematic illustration of the formation of BT nanoparticles with the polymer shell. b) TEM images of BT nanoparticles coated with PS-*b*-PSVBC. c) The dependences of dielectric permittivity on frequency for BT(PS-*b*-PSVBC)/PS and BT(*r*-PSVBC)/PS nanocomposites films. Reproduced with permission.^[90] Copyright 2010, American Chemical Society.

were afterward filled into PS matrix. As shown in **Figure 14a**, the BaTiO₃ nanoparticles were first treated with NaH to form oxy anion on the particle surface, which could react with the chloromethyl group from PS-*b*-PSVBC. In order to obtain to a robust and stable polymer shell, the nonpolar chloromethyl functional blocks of PS-*b*-PSVBC were transformed into polar ammonium blocks through a two-step process. Consequently, a self-assembled and dense layer with thickness of 5–7 nm was formed and completely wrapped the BaTiO₃ nanoparticles (BT(PS-*b*-PSVBC)) (Figure 14b). Such a copolymer layer exhibited strong interaction with BaTiO₃ and also resulted in good dispersion of BaTiO₃ nanoparticles in PS matrix. The outer PS shell of BT(PS-*b*-PSVBC) was found significant to improve the dielectric properties of nanocomposites, e.g., a high dielectric permittivity of 44 was achieved in BT(PS-*b*-PSVBC)/PS composite, while the dielectric permittivity decreased to 22 for the nanocomposites filled with random PSVBC-grafted BaTiO₃ nanoparticles (BT(*r*-PSVBC)) without outer PS shell (Figure 14c). Moreover, the none-charged layer of PS in BT(PS-*b*-PSVBC) could also decrease the exposure of inner charged PSVBC surface, thus giving rise to low leakage current and high breakdown strength in nanocomposites. The nanocomposites showed a high theoretical energy density of 9.7 J cm⁻³.

Different from “grafting to” approach, when preparing core-shell-structured nanocomposites from “grafting from” route, the polymer shell is produced via an in situ polymerization of monomers on the initiated nanoparticle surface. Thus, the introduction of initiating sites on to the particle surface is necessary for such approach. Marks and co-workers^[196–198] reported a “grafting from” method for in situ synthesis of polyolefin-based nanocomposites by using high-ε_r ceramic nanoparticles as fillers, including BaTiO₃, TiO₂, and ZrO₂. A type of metallocene precatalyst [ac-ethylenebisindenyl] zirconium dichloride (BIZrCl₂) was chemisorbed on the surface of nanoparticles. In situ polymerization of propylene was activated by the

catalysts preanchored on the particle surface, then resulting in nanocomposites with well-dispersed nanoparticles in polyolefin matrix. Such 0–3 nanocomposites exhibited high dielectric constant, high breakdown strength, and good energy storage properties. Based on this approach, they extended the in situ method from ceramic to oxide-coated metallic nanoparticles, and achieved a high recoverable energy density in metallic aluminum–polypropylene nanocomposites.^[199] The precatalyst BIZrCl₂ was also introduced to the native Al₂O₃ surface of metallic Al nanoparticles, followed with exposure to propylene, affording 0–3 metal isotactic polypropylene nanocomposites (**Figure 15a**). The nanocomposites were of high quality with no microstructural defects, the maximum dielectric constant was achieved near the percolation threshold of 0.16 vol%. As shown in **Figure 15b,c**, these nanocomposites exhibited excellent dielectric and energy storage capabilities with low leakage current densities of 10⁻⁷–10⁻⁹ A cm⁻² at an electric field of 10⁵ V cm⁻¹, and low dielectric loss in the 100 Hz–1 MHz frequency range. The best performing nanocomposite was achieved at 10% volume fraction of Al nanoparticles, yielding a recoverable energy storage density as high as 14.4 J cm⁻³.

Also through a “grafting from” approach, core-shell-structured BaTiO₃/polymethylethacrylate (PMMA) nanocomposites were prepared with the outer PMMA polymer shell directly serving as the matrix.^[200,201] The in situ polymerization of MMA monomers on the particle surface enabled controlling the thickness of the formed polymer shell, thus the concentration of BaTiO₃ nanoparticles as well as dielectric constant of nanocomposites could be adjusted. Instead of serving as the matrix as the PMMA shell mentioned above, other two types of fluoroalkyl acrylate monomers with different molecular structures were also grafted onto the BaTiO₃ surface via surface-initiated polymerization to modify the interface structures in the P(VDF-HFP)/BaTiO₃ nanocomposites.^[202] As illustrated in **Figure 16**, the polymer shell synthesized from fluoroalkyl

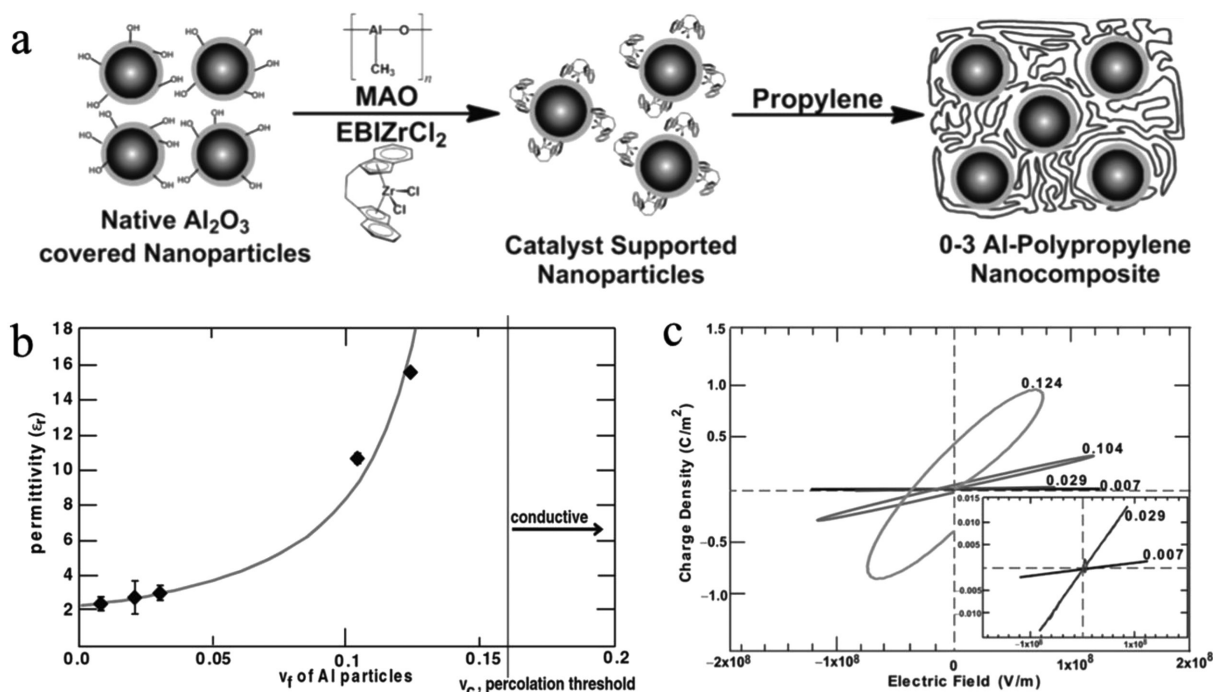


Figure 15. a) Schematic synthesis process of 0–3 polypropylene nanocomposites. b) Dielectric permittivity for Al/iso-PP nanocomposites as a function of Al nanoparticle volume fraction. c) The measured charge density–electric field loops for Al/iso-PP nanocomposites having the indicated nanoparticle volume fractions. The inset shows the lower volume fraction polarizations in an expanded scale. Reproduced with permission.^[199]

acrylate monomers with long side groups 1H,1H,2H,2H heptafluorodecyl acrylate (HFDA) resulted in a compact surface structure because of low chain mobility and ordered structure. The compact surface consequently weakened the interaction between the nanoparticles and polymer matrix. In contrast, the low chain mobility and disordered structure of polymer shell from monomers with short side groups tri-fluoroethyl acrylate (TFEA) led to a loose surface, which tends to form a compact interface with polymer matrix via strong interaction. Owing to the compact interface structure,

P(VDF-HFP)/BT–PTFEA nanocomposite exhibited much lower dielectric loss than P(VDF-HFP)/BT–PHFDA nanocomposite. Moreover, the strong interaction between nanoparticles and matrix could restrict the mobility of polymer chains, thus suppressing the dipole movement of polymer chains and mobile charges, which is also in favor of enhanced dielectric breakdown behaviors.

Homogeneous dispersion of nanoparticles in the matrix is the premise of fully realizing the synergistic combination of advantages of polymer and fillers. In addition, the enhancement

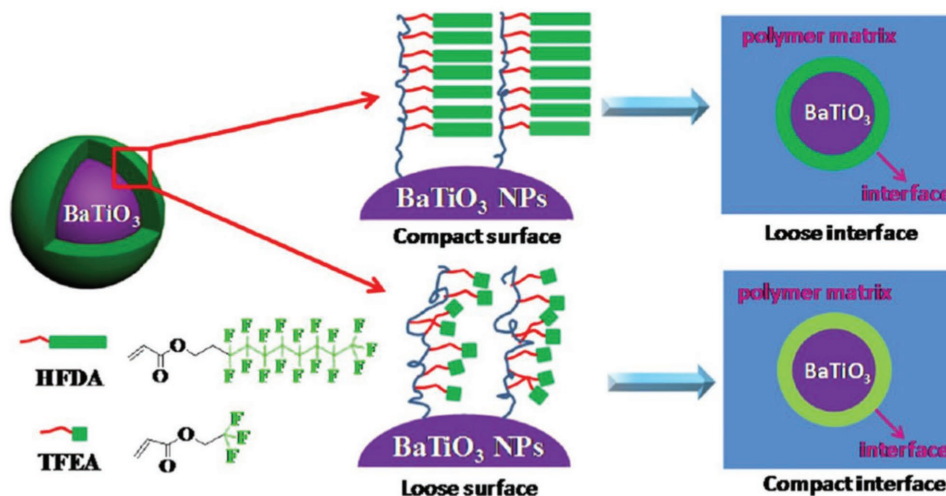


Figure 16. Schematic illustration of the fluoropolymer@BT nanoparticles and the interfacial regions within the nanocomposites. Reproduced with permission.^[202] Copyright 2013, American Chemical Society.

of local electrical and dielectric properties at interface region also contributes enhancement of energy storage performances in nanocomposites, as discussed in the next part.

5. Interfacial Enhancement Effect of Breakdown Strength and Dielectric Response

The local dielectric and breakdown properties of the interface regions are different from the bulk properties of both polymer and ceramic fillers (see Figure 3c), and can be amplified by the significant volume proportion of interfaces. Therefore, by regulation of the local ceramics/polymer interfacial electrical and dielectric characteristics, e.g., interface electronic states, interfacial polarization, dielectric and electrical mismatch, it is feasible to obtain much enhanced dielectric and energy storage properties in nanocomposites beyond the simple addition of unitary polymer and ceramic fillers.

5.1. Interfacial Enhancement of Breakdown Strength

Dielectric breakdown is quite related with the mobile charges and conductivity inside the nanocomposites. The use of chemical modifier enables not only enhancement of the compatibility and nanoparticles dispersion, but also improving the local electrical behaviors at interfaces for reduced leakage current and increased breakdown strength by their electronic nature.^[203–206] The chemically bonded molecules on the ceramic nanoparticle surface can influence the electronic states of interfaces, hence regulating the electron trapping, scattering, and even local conductivity.^[207,208] Schuman and co-workers^[209] demonstrated that proper selection of modifier ligands with regulated functional groups and electronic structures can contribute to improved interfacial electrical behaviors. Five kinds of organophosphate ligands including phenyl phosphate (PP), aminophenyl phosphate (APP), nitrophenyl phosphate (NPP), chlorophenyl phosphate (CPP), and aminoethyl phosphate (AEP) were used to modify the surface of TiO₂. As illustrated in Figure 17a,b, in the molecular structures of those ligands, both a phosphate group and a terminal functional group were attached to a benzene ring. Those terminal functional groups exhibited either electron donating or electron withdrawing characteristic. According to the dielectric tests, using electropositive aromatic organophosphate ligands with an electron withdrawing functional group, such as nitro, to modify the TiO₂ surface, improved leakage current and low dielectric loss were achieved, while the ligands with electron donating groups were not in favor for such enhancement. A Hammett linear free energy relationship was used to correlate the interface polarity and electron delocalization to the dielectric properties of nanocomposites, and the advantage of electropositive ligands in minimizing leakage current and dielectric loss could be characterized with a linear Hammett relationship against group polarity effects. Diebold et al.^[210] also studied the effects of silane coupling agent chemistry on breakdown properties when they are bonded on the silicon oxide surface. Various silane coupling agents with different molecular structures were deposited on the native silicon oxide, and the breakdown strengths of the silane coupling agents/silicon oxide hybrid dielectrics were measured with conductive atomic force microscopy. It was found that silane coupling agent with fluorinated groups can withstand higher breakdown strength than the hydrocarbon counterparts. Two possible factors accounted for the enhancement. First, the fluorinated silane coupling agent could suppress the tunneling current at high electric field. Second, the highly electronegative nature of fluorine and ionic C–F bond may act as electronic scattering points and traps, both resulting in suppressed local conductivity and improved breakdown strength.

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5.2. Interfacial Enhancement of Dielectric Response

Two important interface characteristics have direct bearing on the dielectric response in ceramics/polymer nanocomposites. First, the polymer chains have different packing density, configuration, crystallinity, and free volume at interface regions, thus alternating the dielectric constant of nanocomposites.^[26,154,211] Second, the effect of space-charge polarization at interfaces (interfacial polarization) on frequency dependent dielectric properties is significant.^[161,170,212] Modification of interfaces enables achieving high dielectric constant in nanocomposites beyond a simply combined value of polymer and filler.

The dielectric enhancement by interface and its contribution to the composite have been identified by some theoretical works. Different from standard two-phase mixture models such as the famous Maxwell–Garnett formula,^[213,214] in these interface-involved theoretical models, the interface is usually considered as a separate phase (“interphase”) in addition to polymer and fillers.^[215–219] For example, Todd and Shi^[220] studied the complex permittivity of composite systems by an interphase power-law model, which was constructed on the permittivity and volume fractions of matrix, filler, and interphase between them. The interphase power-law model revealed that the effective permittivity of composite system is significantly affected by the interphase permittivity and surface area. The consideration

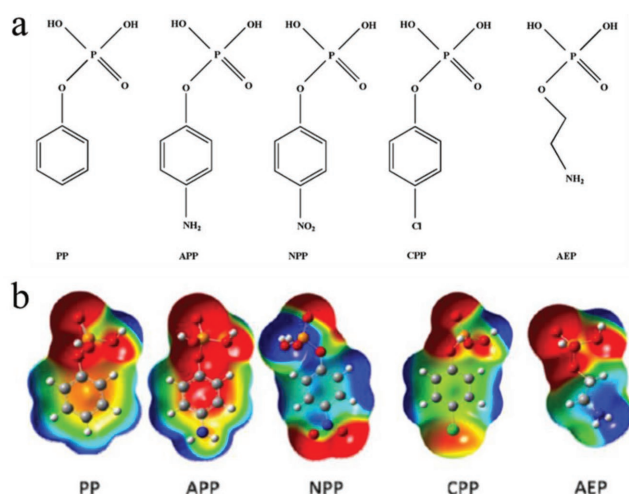


Figure 17. a) Molecular structures and b) DFT calculated electrostatic potential maps of different organophosphate ligands. Reproduced with permission.^[209] Copyright 2013, American Chemical Society.

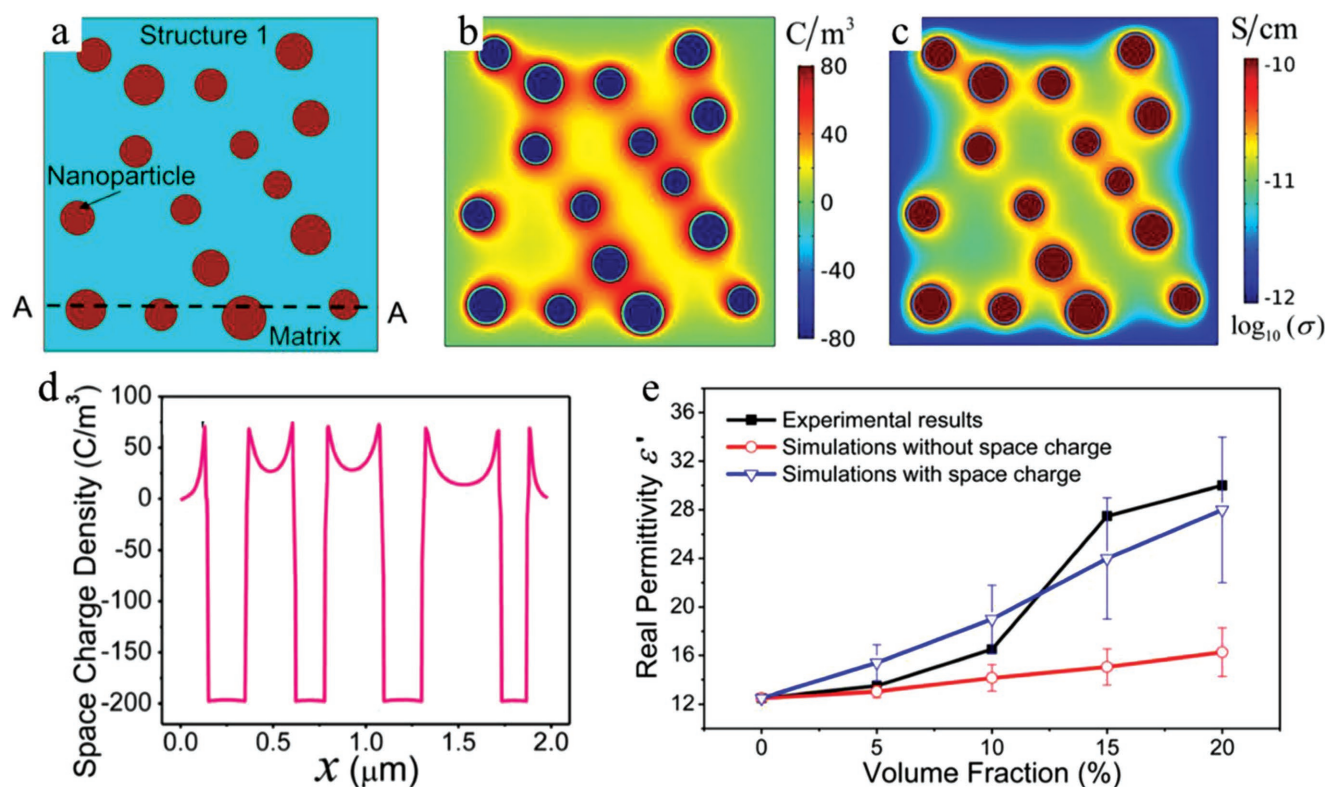


Figure 18. a) Schematic illustration of modeled nanocomposite structure. b) Space charge and c) conductivity distribution inside the nanocomposite. d) Cross-section profile along A–A. e) Variation of experimental and simulated dielectric permittivity on the volume fraction of nanoparticles. Reproduced with permission.^[170] Copyright 2017, American Institute of Physics.

of interface region effects enables this model with better prediction of composite permittivity than the standard two-phase mixture models. Vo and Shi^[216] theoretically investigated the permittivity of polymer/filler composites by taking into account the permittivity ratio between polymer and filler and the interaction strength between them. In this model, Maxwell–Wagner formulation was included and demonstrated as an asymptotic limit. With consideration of the existence of interphase, the modeled dependence of effective composite permittivity on the filler concentration was nonmonotonic with an extreme. Beyond effective medium theory–based mixture rules,^[221,222] the interface effects on the dielectric properties can also be simulated by phase field method. Shen et al.^[170] studied the space-charge effects on dielectric responses of polymer nanocomposites by using the phase field method. In the modeled system (Figure 18a), SrTiO_3 nanoparticles were dispersed in the polyethylene matrix, and the ion concentration of nanocomposite was fixed at 10^{13} cm^{-3} . The distribution of space charges is shown in Figure 18b,d. When the nanoparticles were negatively charged in the nanocomposites, the positively charged ions were concentrated at the interfaces and gradually reduced into the bulk matrix, which agreed with the electric double layer model. The high density of charges also resulted in high local conductivity, as shown in Figure 18c. Based on this model, the dielectric constants of nanocomposites with different volume fractions of nanoparticles were calculated. It was found that the dielectric permittivity exhibited slow increase with increasing filler volume fraction when the space-charge contribution was not considered. While

after the addition of space-charge effects, the dielectric permittivity increased much faster, which was more in agreement with the experimental measured results. The computational results demonstrated that the dielectric response is largely determined by the space charges accumulated at interfaces.

However, the enhancement of dielectric constant from space-charge polarization is sometimes against the enhancement of energy storage performances. The polarized interfacial charges may also increase the local electric field and also create AC conduction, resulting in decreased breakdown strength and increased dielectric loss.^[223] Therefore, dielectric enhancement with suppressed dielectric loss and increased breakdown strength are highly desired. Recently, we demonstrated that by constructing a hierarchical interface structure in the fiber reinforced nanocomposites, the dielectric constant as well as electrical displacement can be effectively improved by regulating atomic-scale interface structure within the nanofillers, meanwhile, the dielectric loss was maintained at a low level. More importantly, the breakdown strength could also be improved, thus giving rise to concurrent enhancement of ϵ_r and E_b , leading to much elevated energy density.^[44,148,224] Such hierarchical interface structure relied on a large-aspect-ratio composite nanofibers $\text{BaTiO}_3@\text{TiO}_2$ (BTO@TO_nfs) by embedding BaTiO_3 nanoparticles (BTO_nps) into TiO_2 nanofibers (TO_nfs) (Figure 19a,b). These BTO@TO_nfs were afterward filled into PVDF matrix, and nanocomposite films were obtained through a solution casting method. In this nanocomposite, more interfaces, i.e., those between BaTiO_3 and TiO_2 , could be introduced

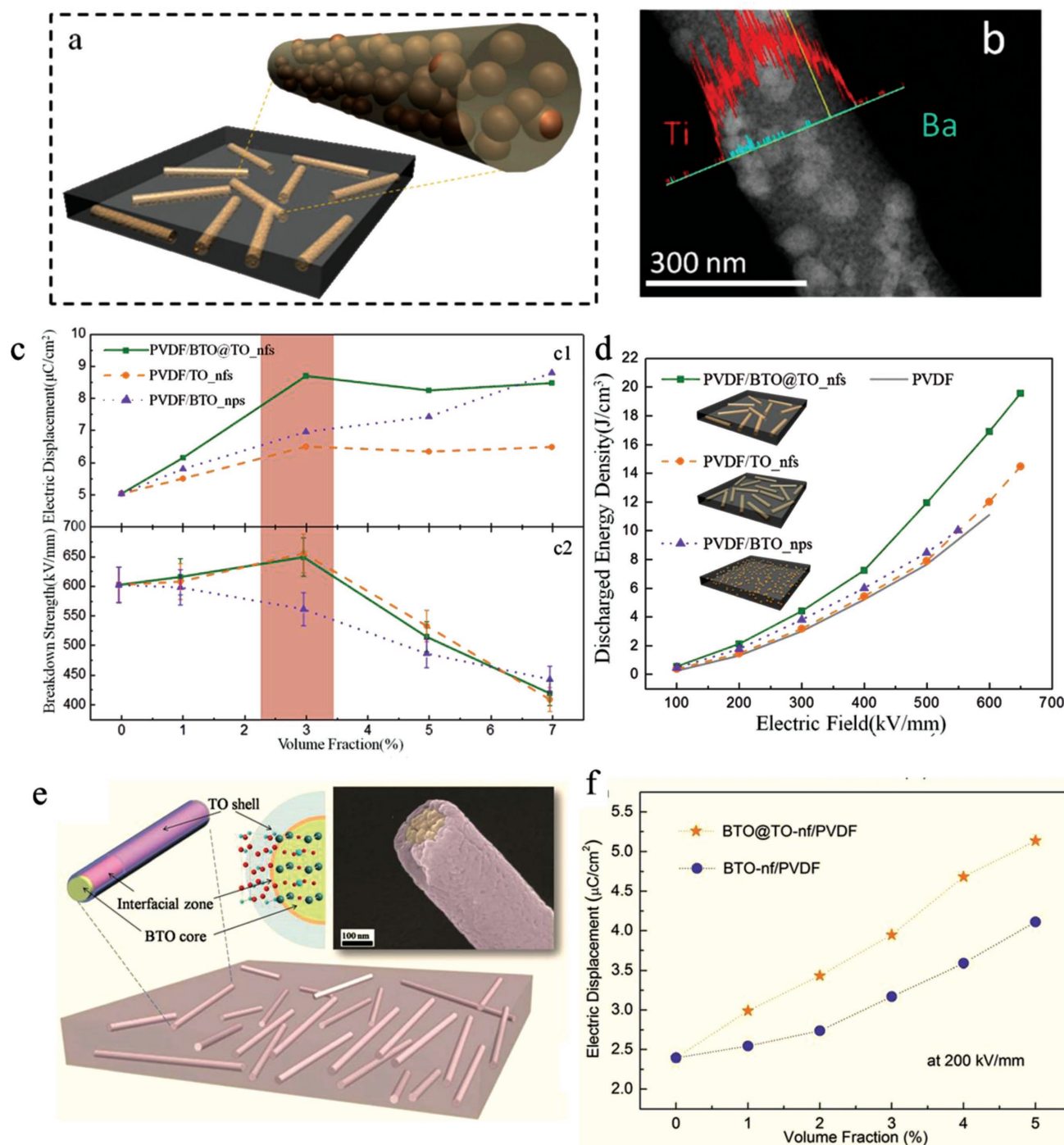


Figure 19. a) Schematic illustration of nanocomposites with hierarchical interface structure. b) TEM image with line mapping of BTO@TO_nfs. c) The variation of electric displacement and breakdown strength with increasing the volume fraction of nanofillers for different nanocomposites. d) Energy density as a function of the electric field for different nanocomposites. a–d) Reproduced with permission.^[44] e) Schematic illustration of PVDF-based nanocomposites filled with core-shell-structured BTO@TO nanofibers. f) Variation of electric displacement with the volume fraction of nanofillers for nanocomposite filled with core-shell BTO@TO_nfs and BTO_nfs. e, f) Reproduced with permission.^[226] Copyright 2016, Royal Society of Chemistry.

in addition to fiber/polymer interfaces without increasing the total volume fraction of inorganic fillers. The dielectric tests demonstrated that the interfaces inside the BTO@TO_nfs could contribute to significant interfacial dielectric enhancement, resulting in a high ϵ_r (≈ 41) in polymer nanocomposites

with a low volume fraction ($\approx 10\%$) of nanofillers. Compared with PVDF nanocomposites filled with bare BTO_nfs and TO_nfs, the nanocomposites with hierarchical interfaces exhibited obvious advantages in both dielectric constant and electric displacement. Moreover, due to the large aspect ratio

of nanofibers, the breakdown strength of PVDF/BTO@TO_nfs nanocomposites also was improved to 646 MV m^{-1} at 3 vol% of nanofillers, much higher than the pure PVDF ($\approx 599 \text{ MV m}^{-1}$). Thus, concurrently improved electric displacement and breakdown strength could be achieved with optimal volume fraction of BTO@TO_nfs, as shown in Figure 19c. As a result of the concurrent enhancement of E_b and D , an ultrahigh discharged energy density of $\approx 20 \text{ J cm}^{-3}$ was achieved at 646 MV m^{-1} in the PVDF nanocomposite filled with 3 vol% of BTO@TO_nfs (Figure 19d). We further extended the hierarchical interface structure into P(VDF-HFP) nanocomposites. The simultaneous enhancement of E_b and D was also achieved in P(VDF-HFP)/BTO@TO_nfs nanocomposites, and owing to higher intrinsic breakdown strength of the P(VDF-HFP) matrix polymer, an unprecedented giant energy density of $\approx 31.2 \text{ J cm}^{-3}$ was achieved. More importantly, a high discharge efficiency of $\approx 78\%$ was also achieved even at an ultrahigh electric field of 800 MV m^{-1} . The analysis of electrical displacement referring to the specific interfacial area for different nanocomposites indicated that hierarchical interfaces inside the BTO@TO_nfs can induce significant interfacial polarization which is even higher than the interfacial polarization at filler/polymer interfaces. The atomic scale characterization of interface regions revealed that a zonal abnormally structured interfacial region with thickness of 3–4 nm existed around BTO nanoparticle, inside of which Ba^{2+} and Ti^{4+} were mutually occupied. The mutual occupation of Ba–Ti accounted for the enhanced interfacial polarization, which has been further identified by the ab initio DFT calculation results. This work proposed an effective approach to simultaneously enhance the dielectric constant and breakdown strength of polymer nanocomposites by interface engineering. A similar hierarchical structure has been also constructed in $\text{BaTiO}_3/\text{Al}_2\text{O}_3$ nanofiber by embedding BaTiO_3 nanoparticles inside the Al_2O_3 nanofiber, the $\text{BaTiO}_3/\text{Al}_2\text{O}_3$ hierarchical structure resulted in a high energy density of 10.58 J cm^{-3} in the PVDF-based nanocomposite.^[225]

Similar enhancement could also be achieved in nanocomposites filled with core-shell-structured $\text{BaTiO}_3/\text{TiO}_2$ nanofibers reported by Hu and co-workers^[226] (Figure 19e). The composite fiber was fabricated by coaxial electrospinning of BaTiO_3 gel and TiO_2 gel. After embedding those nanofibers into PVDF matrix, the charge movements in the interfacial region within the nanofiber enabled producing of additional polarization, thus resulting in high dielectric constant as well as electrical displacement (Figure 19f). Meanwhile, the dielectric breakdown strength was also improved by restricting the mobile charges within the interfacial zone other than forming a percolation path in the matrix. Consequently, a high energy density of 10.94 J cm^{-3} was reached at electric field of 360 MV m^{-1} by concurrently improved dielectric constant and breakdown strength at the filler volume fraction of 3%.

The interfacial dielectric response on the polymer side is usually associated with the structure characteristics of polymer molecular chains, e.g., free volume fraction, mobility, crystallinity, and configurations. As a matter of fact, the dielectric constant of nanocomposites does not necessarily improve with application of fillers, as indicated in some epoxy nanocomposites.^[227] The dielectric constant on the polymer side at the interfacial region is sometimes supposed to be decreased due to

the reduced mobility and the low packing density of polymer chains at the crude polymer/filler interfaces.^[26,169,174] Nevertheless, it has been demonstrated that the polymer side-contributed enhancement of dielectric constant could also be achieved by proper interfacial control. For example, Zhang and co-workers^[228] recently prepared polyetherimide (PEI) nanocomposites filled with low dielectric constant fillers. These fillers, e.g., Al_2O_3 , MgO , SiO_2 , and BN , have similar dielectric constant with PEI matrix, whereas the dielectric constant of nanocomposites was significantly increased at very low volume fraction of nanoparticles. The enhancement of dielectric constant was ascribed to the reduced constraints for dipole reorientations on the polymer side in interfacial regions, which has been evidenced by the reduced T_g and amorphous halo of X-ray diffraction (XRD) pattern in the PEI nanocomposites. However, such interfacial enhancement seems to be only applicable for polar polymer-based nanocomposites such as polyetherimide and polyimide, and nonpolar polymers such as polystyrene do not show this dielectric enhancement. Similar enhancement of dielectric constant was also observed in PMMA-based nanocomposites by using nanosized TiO_2 as fillers.^[229] The stiff chains and high cohesion energy density of PMMA chains resulted in loosely packed interface layer, which enabled high free volume for enhanced segmental motion of chains as well as increased orientation of polar side chain groups, leading to increased dielectric constant. The dielectric constant on the polymer side can also be improved by inducing more polar phase of polymers in the nanocomposites. For example, $-\text{C}_2\text{H}_4(\text{CF}_2)_6\text{F}$ (POTS) chains was used as surface modifier in the P(VDF-HFP)/ SiO_2 nanocomposites to generate more polar phase of P(VDF-HFP). The gap between adjacent POTS chains on the particle surface could be regulated by controlling the surface density of POTS, and a proper gap (6.42 Å) enabled selective formation of all-*trans* TTTT configuration of P(VDF-HFP) chains but repelling chains of TGTG configuration due to their differences in steric hindrance. The TTTT configuration finally resulted in more β phase of P(VDF-HFP) as well as improved dielectric constant of nanocomposites.^[211]

5.3. Manipulating the Interface Dielectric Mismatch

The dielectric mismatch exists at interface due to the large difference of dielectric constant and electrical conductivity between high- ϵ_r ceramic nanoparticles and low- ϵ_r polymers (the difference usually ranges over 2–3 orders), consequently, the charges and electric field will be redistributed in the heterogeneous nanocomposites under a voltage applied. The dielectric mismatch leads to two adverse effects: first, there would be strong electric field distortion existing across the interface region, and these electric field intensely concentrated regions are most vulnerable for the electrical damage and tend to generate the breakdown of nanocomposites;^[167,230] second, the charges accumulated at interfaces induce strong interfacial polarization according to the Maxwell–Wagner theory,^[231] resulting in high dielectric loss and leakage current. In order to avoid dielectric mismatch, choosing polymer and nanofiller with similar dielectric constant is feasible. For example, in the P(VDF-TrFE-CFE)/ TiO_2 nanocomposites,^[106] the matrix polymer and nanofiller have comparable dielectric constant of

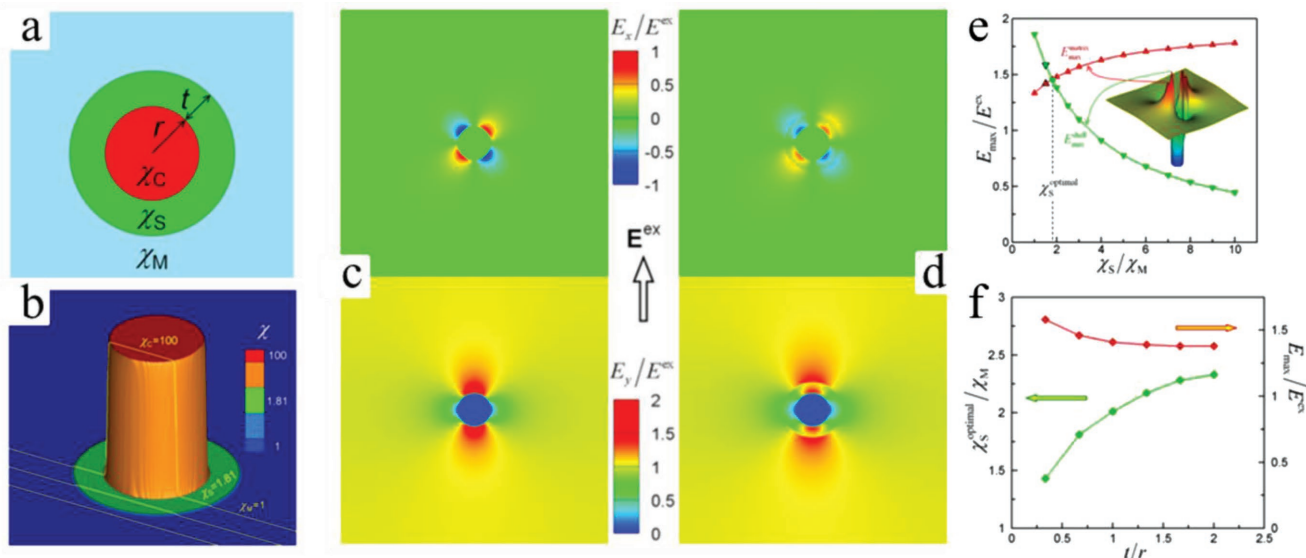


Figure 20. a) Schematic illustration of composite filled with core-shell-structured nanoparticle and the design variables. b) 3D view of dielectric distribution in modeled composite. The simulated electric field distribution in composite for particle c) without core-shell structure and d) with core-shell structure, the electric field components of E_x and E_y are shown in top and bottom parts, respectively. The effects of e) shell dielectric susceptibility (χ_s/χ_m) and f) shell thickness on the electric field concentration in composites. Reproduced with permission.^[230] Copyright 2011, American Institute of Physics.

42 and 47, respectively. The nanocomposite showed no increment in dielectric constant with addition of TiO_2 nanoparticles at low electric fields, however, it exhibited excellent high field dielectric performances with much enhanced electric displacement and energy density.

Another approach to manipulating the dielectric mismatch at interface is to construct dielectric gradients between high- ϵ_r and low- ϵ_r phases. Usually, this can be realized by core-shell structures of nanofillers. Wang et al.^[230] studied the electric field distribution behaviors in polymer nanocomposites filled with high- ϵ_r filler by phase field simulations. In this simulation, by setting the dielectric susceptibility of core (χ_c), shell (χ_s), and matrix (χ_m) from high to low, a dielectric gradient could be achieved (Figure 20a,b). Then, the distributions of electric field in dielectric composites were simulated for filler particles with and without core-shell structure. As shown in Figure 20c,d, the results demonstrated that core-shell structure can significantly reduce the maximum value of local electric field in composites in both x - and y -directions, thus mitigating the electric field concentration and distortion at the interface region. In order to optimize and fully exploit the advantage of core-shell structure, they further investigated the effects of shell thickness and shell dielectric constant on tuning the electric field distribution. The optimal dielectric susceptibility was confirmed as $\chi_s/\chi_m = 1.81:1$, and the dielectric susceptibility of core was found has little effect on determining the optimal value. Moreover, a thicker shell was preferred to minimize the electric field concentration. This work theoretically identified the validity of core-shell structure in mitigating the field concentration and improving the dielectric breakdown strength in polymer nanocomposites, and also offered guidance for selecting the appropriate materials.

Such a core-shell structure approach has been widely used in experimental, and various buffer layer with lower dielectric

constant, e.g., Al_2O_3 , SiO_2 , TiO_2 , have been coated onto the high dielectric constant core.^[232–243] For example, Marks and co-workers^[197,198] reported a layer-by-layer method to coat Al_2O_3 shells on the surface of high- ϵ_r ceramic nanoparticles including BaTiO_3 , ZrO_2 , MgO , SrTiO_3 , and $\text{Ba}_{0.5}\text{Sr}_{0.5}\text{TiO}_3$. As illustrated in Figure 21a, the core-shell structure was achieved by ambient exposure of the methylalumoxane (MAO) cocatalyst/activator coating in the air, thus the MAO coating could be oxidized and hydrolyzed in this procedure to form a layer of amorphous Al_2O_3 . These core-shell nanoparticles were afterward incorporated into isotactic polypropylene matrix, in which the Al_2O_3 ($\epsilon_r \approx 10$) shell could serve as dielectric buffer between high- ϵ_r ceramic nanoparticles ($\epsilon_r \approx 50\text{--}2000$) and low- ϵ_r matrix ($\epsilon_r \approx 2$). The thickness of Al_2O_3 shell ranged over 2–10 nm by controlling the number of layers from 1 to 5. The leakage and dielectric loss of the nanocomposites were greatly reduced with the presence of Al_2O_3 shell, contributing to high energy efficiency and breakdown strength. According to the Maxwell–Garnett effective medium theory calculated results, the nanocomposites with thicker Al_2O_3 layer (10 nm) possessed slightly higher dielectric constant, while significantly lower dielectric loss (Figure 21b,c). Similar enhancement could also be achieved by using SiO_2 layer as the dielectric buffer. Wang and co-workers^[244] reported PVDF-based nanocomposites by using core-shell-structured $\text{BaTiO}_3/\text{SiO}_2$ nanoparticles as fillers. The formation of SiO_2 layer could reduce the dielectric loss by suppressing the Maxwell–Wagner interfacial polarization. Compared to the nanocomposites filled with untreated BaTiO_3 nanoparticles, the nanocomposites filled with core-shell-structured nanoparticles exhibited slimmer D – E loops with both suppressed maximum polarization (14% of reduction) and remnant polarization (53% of reduction), which resulted in improved energy efficiency and energy density.

Such dielectric gradient with core-shell structure can also be constructed for nanofibers.^[245] Zhai and co-workers^[95]

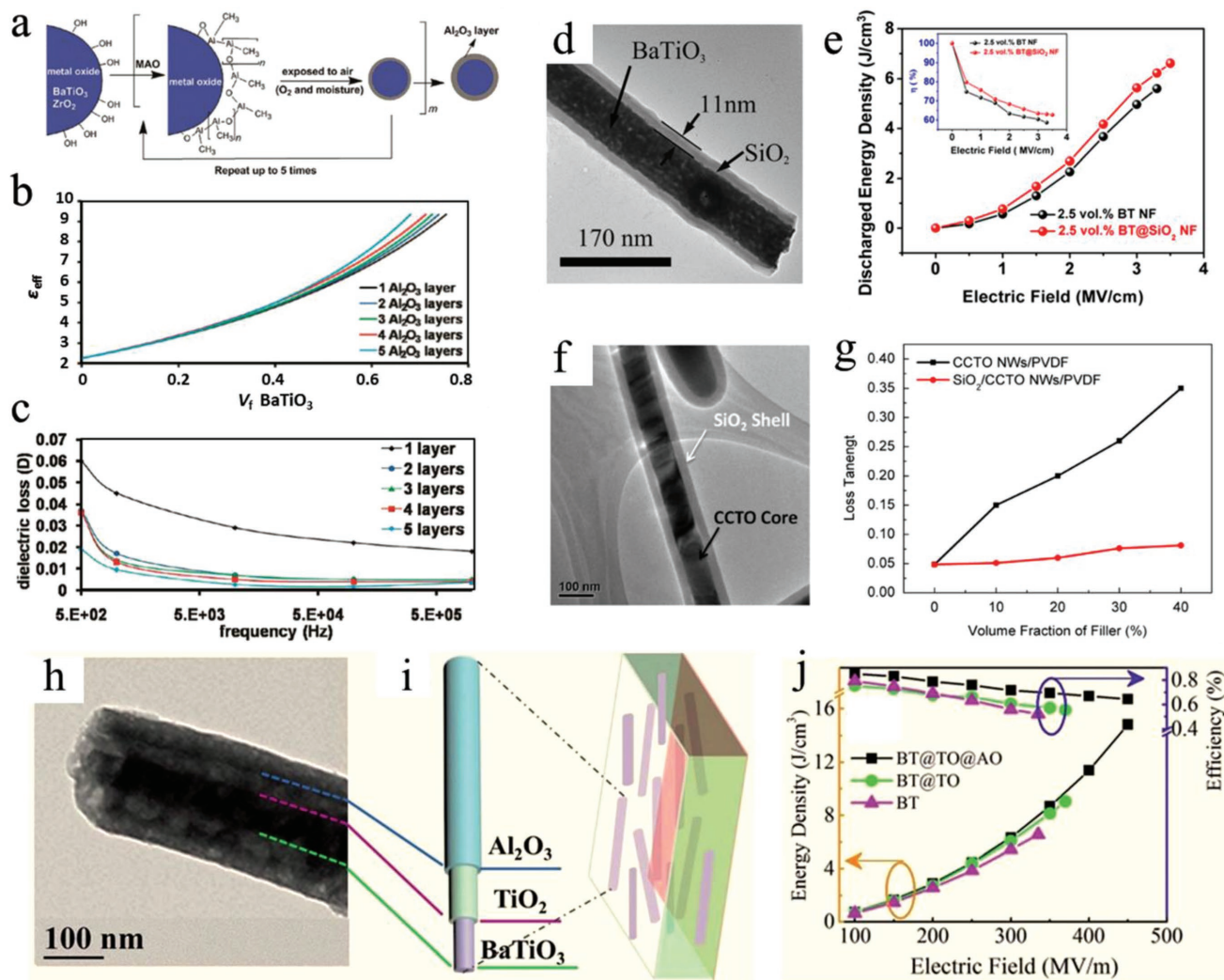


Figure 21. a) Schematic illustration of coating Al_2O_3 layers onto ceramic nanoparticles surface. b) The dependent of Maxwell–Garnett theory calculated effective permittivity on the volume fraction of the BaTiO_3 nanoparticles with different Al_2O_3 thicknesses (the nanoparticle volume fractions are 7%). a–c) Reproduced with permission.^[197] Copyright 2010, American Chemical Society. d) TEM image of core–shell-structured BaTiO_3 @ SiO_2 nanofibers. e) The energy density of nanocomposites filled with BaTiO_3 @ SiO_2 nanofibers and BTO nanofibers as a function of electric field. d,e) Reproduced with permission.^[95] Copyright 2015, American Institute of Physics. f) TEM image of core–shell-structured CCTO@ SiO_2 nanowires. g) The dielectric loss of nanocomposites filled with CCTO@ SiO_2 nanowires and CCTO nanowires as a function of filler volume fraction. f,g) Reproduced with permission.^[123] Copyright 2015, Elsevier. h) TEM image of core–double shell-structured BaTiO_3 @ TiO_2 @ Al_2O_3 nanofiber. i) Schematic illustration of nanocomposites filled with nanofibers. j) Energy density and energy efficiency of nanocomposites filled with different nanofibers. h–j) Reproduced with permission.^[246] Copyright 2017, Royal Society of Chemistry.

fabricated core–shell-structured BaTiO_3 @ SiO_2 nanofibers (Figure 21d), the BaTiO_3 nanofibers were presynthesized by electrospinning, and SiO_2 layer was subsequently coated onto the fiber surface by the hydrolysis reaction of tetraethoxysilane (TEOS). Similar to the BaTiO_3 @ SiO_2 nanoparticles, the nanocomposites filled with BaTiO_3 @ SiO_2 nanofibers also showed decreased energy loss and increased discharged energy density compared with bare BaTiO_3 nanofibers. The highly insulating SiO_2 layer has also been employed to handle the dielectric loss in polymer nanocomposites filled with calcium copper titanate (CCTO) nanofibers.^[123] CCTO possesses a giant dielectric constant (10^4 – 10^5) as well as relatively high loss tangent. The electric field distortion becomes more

intense when CCTO is used as dielectric reinforcement in polymer nanocomposites. In order to decrease the dielectric loss while maintaining the high dielectric constant, core–shell-structured CCTO@ SiO_2 nanofibers were constructed. CCTO nanofibers were presynthesized with a two-step hydrothermal reaction, afterward, the SiO_2 layer was coated by the hydroxylation of TEOS (Figure 21f). Owing to the intrinsic high dielectric constant of CCTO, the dielectric constant of nanocomposites filled with 40 vol% of CCTO reached around 70, which was almost 7 times higher than the pure PVDF matrix. Compared to the nanocomposites filled with bare CCTO nanofibers, the dielectric loss was significantly reduced with the existence of the SiO_2 layer from CCTO@ SiO_2 , while the high dielectric

constant was still maintained (Figure 21g). Specifically, the loss tangent was decreased from 0.35 for CCTO to 0.081 for CCTO@SiO₂, which was a reduction of 432%, and the dielectric constant exhibited only a slight decrease of 6.8% from 73 to 68. In order to obtain smaller dielectric gradient, Pan et al.^[246] constructed core–double-shell-structured BaTiO₃@TiO₂@Al₂O₃ nanofiber (Figure 21h,i) by three layer coaxial electrospinning technique. The TiO₂ ($\epsilon_r \approx 100$) and Al₂O₃ ($\epsilon_r \approx 10$) shell layer enable gradually varied dielectric constant from the BaTiO₃ core to the PVDF matrix. This core–double-shell structure exhibited more efficiency in reducing the leakage current and dielectric loss compared to core–shell-structured BaTiO₃@TiO₂ nanofiber, leading to higher breakdown strength. Finally, a high energy density of 14.84 J cm⁻³ was obtained in the nanocomposites with 3.6 vol% BaTiO₃@TiO₂@Al₂O₃ nanofiber, which is much higher than the nanocomposites filled with bare BTO fibers and single shell BaTiO₃@TiO₂ nanofiber.

6. Summary and Outlook

Dielectric capacitors are critical components in advanced electronic and electric power systems, due to their typical capability of delivering an ultrahigh power density, low loss, and withstanding high operating voltage than do other energy storage devices. However, the energy density is rather limited in common unitary polymeric and ceramic dielectric materials by the paradox between dielectric constant and breakdown strength. Polymer nanocomposites by integration of high- E_b polymer matrix and high- $D(\epsilon_r)$ ceramic fillers have become main strategies to overcome the E_b – $D(\epsilon_r)$ trade-off to achieve much elevated energy density. Among the various factors influencing the dielectric and energy storage properties of polymer nanocomposites, the filler–matrix interfaces play a dominant role when the fillers are of nanoscale. Two important features of interfaces, specifically, the significant content of interfaces region and the abnormal interfacial properties, offer the possibility to achieve enhanced dielectric performances in nanocomposites beyond the simply combined properties of polymer and ceramic fillers. Tremendous effects have been made to understand the influence of interfaces on dielectric and breakdown behaviors in polymer nanocomposites, and also to propose effective approaches to modify the polymer/ceramics filler interfaces for improved energy storage performances. The first scope of interface strategy is to improve the organic/inorganic interfacial compatibility for homogeneous dispersion of nanoparticles in polymer matrix, which is the premise of fully realizing the synergistic combination of advantages of polymer and fillers. Surface hydroxylation of nanoparticles, chemical treatment with surface modifiers, grafting of polymer through a polymerization process have been identified as effective approaches to enhance the dispersion of ceramic nanoparticles. The other scope of interface strategy is to regulate the local interfacial characteristics, e.g., interfacial electron trapping or scattering, electronic states, interfacial polarization, redistribution of electric field, charge density, and charge accumulation, to obtain improved dielectric response and breakdown behaviors at interfaces region. The enhancement of properties

at interfaces would be amplified by the significant content of interfaces volume and finally contributes to the enhancement of energy storage properties in dielectric nanocomposites.

The validity of interface enhancement on improving the dielectric and energy storage properties of ceramics/polymer nanocomposites has been identified by various structural and functional targeted approaches. However, challenges are still existing in this interfacial strategy and more efforts are needed. Some present problems and possible solutions are proposed, including: 1) the exact and comprehensive understanding of interface effects on dielectric properties still needs to be further clarified, especially by means of nanoscale, direct, position-dependent detection of local electrical and dielectric response at interface region; 2) dielectric mismatch at interfaces greatly deteriorates the energy storage performances and needs to be mitigated, constructing dielectric gradients across interface is considered as an effective solution to this issue; 3) constructing novel interface structures beyond the conventional polymer/ceramics ones, e.g., hierarchical interface by introducing subinterface inside the nanofillers, have shown great promise to achieve much elevated energy density in polymer nanocomposites.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

dielectric, energy storage, interfaces, nanoparticles, polymer nanocomposites

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- [1] Q. Li, L. Chen, M. R. Gadinski, S. Zhang, G. Zhang, H. Li, A. Haque, L. Chen, T. Jackson, Q. Wang, *Nature* **2015**, 523, 576.
- [2] Z. Dang, J. Yuan, S. Yao, R. Liao, *Adv. Mater.* **2013**, 25, 6334.
- [3] Prateek, V. K. Thakur, R. K. Gupta, *Chem. Rev.* **2016**, 116, 4260.
- [4] Q. Chen, Y. Shen, S. Zhang, Q. M. Zhang, *Annu. Rev. Mater. Res.* **2015**, 4, 433.
- [5] I. Plesa, P. V. Notinger, S. Schloegl, C. Sumereder, M. Muhr, *Polymers* **2016**, 8, 173.
- [6] X. Huang, P. Jiang, *Adv. Mater.* **2015**, 27, 546.
- [7] Q. Wang, L. Zhu, *J. Polym. Sci., Part B: Polym. Phys.* **2011**, 49, 1421.
- [8] P. Barber, S. Balasubramanian, Y. Anguchamy, S. Gong, A. Wibowo, H. Gao, H. J. Ploehn, H. Zur Loye, *Materials* **2009**, 2, 1697.

- [9] E. J. Barshaw, J. White, M. J. Chait, J. B. Cornette, J. Bustamante, F. Folli, D. Biltchick, G. Borelli, G. Picci, M. Rabuffi, *IEEE Trans. Magn.* **2007**, 43, 223.
- [10] C. A. Howells, *Energy Convers. Manage.* **2009**, 50, 1847.
- [11] L. Lebrun, D. Guyomar, B. Guiffard, P. J. Cottinet, C. Putson, *Sens. Actuators, A* **2009**, 153, 251.
- [12] M. Yoon, I. Mahmud, S. Ur, *Ceram. Int.* **2013**, 39, 8581.
- [13] M. Choi, Y. Kim, C. Ha, *Prog. Polym. Sci.* **2008**, 33, 581.
- [14] J. Jang, Y. Song, D. Yoo, T. Kim, S. Jung, S. Hong, J. Lee, T. Lee, *Org. Electron.* **2014**, 15, 2822.
- [15] K. Wolff, U. Hillerlingmann, *Solid-State Electron.* **2011**, 62, 110.
- [16] M. A. Alam, M. H. Azarian, M. G. Pecht, *J. Electron. Mater.* **2012**, 41, 2286.
- [17] A. Burke, *Electrochim. Acta* **2007**, 53, 1083.
- [18] S. Ogitani, S. A. Bidstrup-Allen, P. Kohl, *IEEE/CPMT Int. Electron. Manuf. Technol. Symp.* **1998**, 199.
- [19] J. W. McPherson, J. Kim, A. Shanware, H. Mogul, J. Rodriguez, *IEEE Trans. Electron Devices* **2003**, 50, 1771.
- [20] H. Kishi, Y. Mizuno, H. Chazono, *Jpn. J. Appl. Phys.* **2003**, 42, 1.
- [21] M. Pan, C. A. Randall, *IEEE Electr. Insul. Mag.* **2010**, 26, 44.
- [22] I. Burn, D. M. Smyth, *J. Mater. Sci.* **1972**, 7, 339.
- [23] Y. Wang, X. Zhou, Q. Chen, B. Chu, Q. Zhang, *IEEE Trans. Dielectr. Electr. Insul.* **2010**, 17, 1036.
- [24] T. J. Lewis, *IEEE Trans. Dielectr. Electr. Insul.* **2004**, 11, 739.
- [25] T. J. Lewis, *J. Phys. D: Appl. Phys.* **2005**, 38, 202.
- [26] T. Tanaka, M. Kozako, N. Fuse, Y. Ohki, *IEEE Trans. Dielectr. Electr. Insul.* **2005**, 12, 669.
- [27] Z. Pan, L. Yao, J. Zhai, X. Yao, H. Chen, *Adv. Mater.* **2018**, 1705662, <https://doi.org/10.1002/adma.201705662>.
- [28] L. Zhu, *J. Phys. Chem. Lett.* **2014**, 5, 3677.
- [29] J. Ho, R. Ramprasad, S. Boggs, *IEEE Trans. Dielectr. Electr. Insul.* **2007**, 14, 1295.
- [30] S. J. Laihonon, U. Gavvert, T. Schutte, U. W. Gedde, *IEEE Trans. Dielectr. Electr. Insul.* **2007**, 14, 275.
- [31] M. Rabuffi, G. Picci, *IEEE Trans. Plasma Sci.* **2002**, 30, 1939.
- [32] D. H. Wang, B. A. Kurish, T. Imre, L. Zhu, L. Tan, *J. Polym. Sci., Part A: Polym. Chem.* **2015**, 53, 422.
- [33] S. Wu, Q. Burlingame, Z. Cheng, M. Lin, Q. M. Zhang, *J. Electron. Mater.* **2014**, 43, 4548.
- [34] Y. Wang, X. Zhou, M. Lin, Q. M. Zhang, *Appl. Phys. Lett.* **2009**, 94, 202905.
- [35] S. Wu, W. Li, M. Lin, Q. Burlingame, Q. Chen, A. Payzant, K. Xiao, Q. M. Zhang, *Adv. Mater.* **2013**, 25, 1734.
- [36] S. Wu, M. Lin, Q. Burlingame, Q. M. Zhang, *Appl. Phys. Lett.* **2014**, 104, 72903.
- [37] A. J. Lovinger, *Science* **1983**, 220, 1115.
- [38] L. Zhu, Q. Wang, *Macromolecules* **2012**, 45, 2937.
- [39] J. Gomes, J. S. Nunes, V. Sencadas, S. Lanceros-Mendez, *Smart Mater. Struct.* **2010**, 19, 065010.
- [40] A. Salimi, A. A. Yousefi, *Polym. Test.* **2003**, 22, 699.
- [41] K. Matsushige, K. Nagata, T. Takemura, *Jpn. J. Appl. Phys.* **1978**, 17, 467.
- [42] D. Shah, P. Maiti, E. Gunn, D. F. Schmidt, D. D. Jiang, C. A. Batt, E. P. Giannelis, *Adv. Mater.* **2004**, 16, 1173.
- [43] H. Tang, H. A. Sodano, *Nano Lett.* **2013**, 13, 1373.
- [44] X. Zhang, Y. Shen, Q. Zhang, L. Gu, Y. Hu, J. Du, Y. Lin, C. W. Nan, *Adv. Mater.* **2015**, 27, 819.
- [45] W. Li, Q. Meng, Y. Zheng, Z. Zhang, W. Xia, Z. Xu, *Appl. Phys. Lett.* **2010**, 96, 192905.
- [46] P. Martins, A. C. Lopes, S. Lanceros-Mendez, *Prog. Polym. Sci.* **2014**, 39, 683.
- [47] F. Bauer, E. Fousson, Q. M. Zhang, L. M. Lee, *IEEE Trans. Dielectr. Electr. Insul.* **2004**, 11, 293.
- [48] V. Bharti, H. S. Xu, G. Shanthi, Q. M. Zhang, K. Liang, *J. Appl. Phys.* **2000**, 87, 452.
- [49] V. Bharti, X. Zhao, Q. M. Zhang, T. Romotowski, F. Tito, R. Ting, *Mater. Res. Innovations* **1998**, 2, 57.
- [50] Z. Y. Cheng, Q. M. Zhang, F. B. Bateman, *J. Appl. Phys.* **2002**, 92, 6749.
- [51] B. Daudin, M. Dubus, J. F. Legrand, *J. Appl. Phys.* **1987**, 62, 994.
- [52] J. S. Forsythe, D. Hill, *Prog. Polym. Sci.* **2000**, 25, 101.
- [53] Z. Li, M. D. Arbatti, Z. Y. Cheng, *Macromolecules* **2004**, 37, 79.
- [54] Q. M. Zhang, V. Bharti, X. Zhao, *Science* **1998**, 280, 2101.
- [55] V. Bharti, Q. M. Zhang, *Phys. Rev. B* **2001**, 63, 184103.
- [56] H. Xu, Z. Y. Cheng, D. Olson, T. Mai, Q. M. Zhang, G. Kavarnos, *Appl. Phys. Lett.* **2001**, 78, 2360.
- [57] B. Chu, X. Zhou, K. Ren, B. Neese, M. Lin, Q. Wang, F. Bauer, Q. M. Zhang, *Science* **2006**, 313, 1887.
- [58] B. Chu, X. Zhou, B. Neese, Q. M. Zhang, F. Bauer, *IEEE Trans. Dielectr. Electr. Insul.* **2006**, 13, 1162.
- [59] F. Xia, Z. Y. Cheng, H. S. Xu, H. F. Li, Q. M. Zhang, G. J. Kavarnos, R. Y. Ting, G. Abdul-Sadek, K. D. Belfield, *Adv. Mater.* **2002**, 14, 1574.
- [60] Q. Chen, B. Chu, X. Zhou, Q. M. Zhang, *Appl. Phys. Lett.* **2007**, 91, 62907.
- [61] X. Zhou, B. Chu, B. Neese, M. Lin, Q. M. Zhang, *IEEE Trans. Dielectr. Electr. Insul.* **2007**, 14, 1133.
- [62] F. Guan, L. Yang, J. Wang, B. Guan, K. Han, Q. Wang, L. Zhu, *Adv. Funct. Mater.* **2011**, 21, 3176.
- [63] S. Zhang, C. Zou, D. I. Kushner, X. Zhou Jr., R. J. Orchard, N. Zhang, Q. M. Zhang, *IEEE Trans. Dielectr. Electr. Insul.* **2012**, 19, 1158.
- [64] X. Zhou, X. Zhao, Z. Suo, C. Zou, J. Runt, S. Liu, S. Zhang, Q. M. Zhang, *Appl. Phys. Lett.* **2009**, 94, 162901.
- [65] A. B. Da Silva, M. Arjmand, U. Sundararaj, R. E. S. Bretas, *Polymer* **2014**, 55, 226.
- [66] J. Xu, C. P. Wong, *Appl. Phys. Lett.* **2005**, 87, 82907.
- [67] L. Zhang, W. Wang, X. Wang, P. Bass, Z. Y. Cheng, *Appl. Phys. Lett.* **2013**, 103, 232903.
- [68] Y. Zhang, Y. Wang, Y. Deng, Y. Guo, W. Bi, M. Li, Y. Luo, J. Bai, *Appl. Phys. Lett.* **2012**, 101, 192904.
- [69] A. S. Zeraati, M. Arjmand, U. Sundararaj, *ACS Appl. Mater. Interfaces* **2017**, 9, 14328.
- [70] A. Toor, H. So, A. P. Pisano, *Appl. Surf. Sci.* **2017**, 414, 373.
- [71] S. Moharana, R. N. Mahaling, *Chem. Phys. Lett.* **2017**, 680, 31.
- [72] Z. Chen, H. Li, G. Xie, K. Yang, *RSC Adv.* **2018**, 8, 1.
- [73] F. Du, J. E. Fischer, K. I. Winey, *Phys. Rev. B* **2005**, 72, 121404.
- [74] F. He, S. Lau, H. L. Chan, J. Fan, *Adv. Mater.* **2009**, 21, 710.
- [75] Y. R. Hernandez, A. Gryson, F. M. Blighe, M. Cadek, V. Nicolosi, W. J. Blau, Y. K. Gun Ko, J. N. Coleman, *Scr. Mater.* **2008**, 58, 69.
- [76] J. Li, P. C. Ma, W. S. Chow, C. K. To, B. Z. Tang, J. K. Kim, *Adv. Funct. Mater.* **2007**, 17, 3207.
- [77] J. Y. Yi, G. M. Choi, *J. Electroceram.* **1999**, 3, 361.
- [78] A. K. Das, R. Bhowmik, A. K. Meikap, *AIP Adv.* **2017**, 7, 045110.
- [79] B. Fan, Y. Liu, D. He, J. Bai, *Appl. Phys. Lett.* **2018**, 112, 052902.
- [80] C. Yang, S. Hao, S. Dai, X. Zhang, *Carbon* **2017**, 117, 301.
- [81] M. Yang, H. Zhao, D. He, J. Bai, *Carbon* **2017**, 116, 94.
- [82] W. Ji, H. Deng, Q. Fu, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24, 749.
- [83] X. Zhang, L. Yuan, Q. Guan, G. Liang, A. Gu, *J. Mater. Chem. A* **2017**, 5, 21909.
- [84] Y. Wang, C. Xing, J. Guan, Y. Li, *Polymers* **2017**, 9, 562.
- [85] K. Shehzad, A. Ul-Haq, S. Ahmad, M. Mumtaz, T. Hussain, A. Mujahid, A. T. Shah, M. Y. Choudhry, I. Khokhar, S. Ul-Hassan, F. Nawaz, F. U. Rahman, Y. Butt, M. Pervaiz, *J. Mater. Sci.* **2013**, 48, 3737.
- [86] J. Yuan, Z. Dang, S. Yao, J. Zha, T. Zhou, S. Li, J. Bai, *J. Mater. Chem.* **2010**, 20, 2441.
- [87] Z. M. Dang, Y. H. Lin, C. W. Nan, *Adv. Mater.* **2003**, 15, 1625.

- [88] Z. M. Dang, L. Wang, Y. Yin, Q. Zhang, Q. Q. Lei, *Adv. Mater.* **2007**, 19, 852.
- [89] Y. Shen, Y. Lin, M. Li, C. W. Nan, *Adv. Mater.* **2007**, 19, 1418.
- [90] H. M. Jung, J. Kang, S. Y. Yang, J. C. Won, Y. S. Kim, *Chem. Mater.* **2010**, 22, 450.
- [91] A. Dey, S. De, A. De, S. K. De, *Nanotechnology* **2004**, 15, 1277.
- [92] P. Kim, N. M. Doss, J. P. Tillotson, P. J. Hotchkiss, M. Pan, S. R. Marder, J. Li, J. P. Calame, J. W. Perry, *ACS Nano* **2009**, 3, 2581.
- [93] P. Kim, S. C. Jones, P. J. Hotchkiss, J. N. Haddock, B. Kippelen, S. R. Marder, J. W. Perry, *Adv. Mater.* **2007**, 19, 1001.
- [94] Y. Song, Y. Shen, H. Liu, Y. Lin, M. Li, C. W. Nan, *J. Mater. Chem.* **2012**, 22, 16491.
- [95] S. Liu, S. Xue, B. Shen, J. Zhai, *Appl. Phys. Lett.* **2015**, 107, 32907.
- [96] H. Tang, Z. Zhou, H. A. Sodano, *ACS Appl. Mater. Interfaces* **2014**, 6, 5450.
- [97] H. Tang, Y. Lin, H. A. Sodano, *Adv. Energy Mater.* **2013**, 3, 451.
- [98] L. Xie, X. Huang, B. Li, C. Zhi, T. Tanaka, P. Jiang, *Phys. Chem. Chem. Phys.* **2013**, 15, 17560.
- [99] Z. Dang, Y. Lin, H. Xu, C. Shi, S. Li, J. Bai, *Adv. Funct. Mater.* **2008**, 18, 1509.
- [100] J. Li, J. Claude, L. E. Norena-Franco, S. I. Seok, Q. Wang, *Chem. Mater.* **2008**, 20, 6304.
- [101] Z. Pan, L. Yao, J. Zhai, K. Yang, B. Shen, H. Wang, *ACS Sustainable Chem. Eng.* **2017**, 5, 4707.
- [102] Y. Wang, Y. Hou, Y. Deng, *Compos. Sci. Technol.* **2017**, 145, 71.
- [103] S. Luo, Y. Shen, S. Yu, Y. Wan, W. Liao, R. Sun, C. Wong, *Energy Environ. Sci.* **2017**, 10, 137.
- [104] H. Liu, S. Luo, S. Yu, S. Ding, Y. Shen, R. Sun, C. Wong, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24, 757.
- [105] N. Guo, S. A. DiBenedetto, D. Kwon, L. Wang, M. T. Russell, M. T. Lanagan, A. Facchetti, T. J. Marks, *J. Am. Chem. Soc.* **2007**, 129, 766.
- [106] K. Prabakaran, S. Mohanty, S. K. Nayak, *J. Mater. Sci.: Mater. Electron.* **2014**, 25, 4590.
- [107] E. Tuncer, I. Sauers, D. R. James, A. R. Ellis, M. P. Paranthaman, A. Goyal, K. L. More, *Nanotechnology* **2007**, 18, 325704.
- [108] H. Tang, H. A. Sodano, *Appl. Phys. Lett.* **2013**, 102, 63901.
- [109] J. Zha, G. Chen, Z. Dang, Y. Yin, *J. Electrostat.* **2011**, 69, 255.
- [110] N. Xu, L. Hu, Q. Zhang, X. Xiao, H. Yang, E. Yu, *ACS Appl. Mater. Interfaces* **2015**, 7, 27373.
- [111] Z. Dou, W. Liu, T. Lin, K. Zhou, L. Hang, *Appl. Phys. Lett.* **2017**, 110, 133902.
- [112] N. Xu, Q. Zhang, H. Yang, Y. Xia, Y. Jiang, *Sci. Rep.* **2017**, 7, 43970.
- [113] B. Chu, M. Lin, B. Neese, X. Zhou, Q. Chen, Q. M. Zhang, *Appl. Phys. Lett.* **2007**, 91, 122909.
- [114] B. Chu, M. Lin, B. Neese, Q. Zhang, *J. Appl. Phys.* **2009**, 105, 14103.
- [115] N. Guo, S. A. DiBenedetto, P. Tewari, M. T. Lanagan, M. A. Ratner, T. J. Marks, *Chem. Mater.* **2010**, 22, 1567.
- [116] C. Zou, D. Kushner, S. Zhang, *Appl. Phys. Lett.* **2011**, 98, 82905.
- [117] S. Liu, S. Xue, W. Zhang, J. Zhai, G. Chen, *J. Mater. Chem. A* **2014**, 2, 18040.
- [118] P. Hu, Y. Song, H. Liu, Y. Shen, Y. Lin, C. W. Nan, *J. Mater. Chem. A* **2013**, 1, 1688.
- [119] S. Liu, J. Zhai, J. Wang, S. Xue, W. Zhang, *ACS Appl. Mater. Interfaces* **2014**, 6, 1533.
- [120] P. Wu, M. Zhang, H. Wang, H. Tang, P. Bass, L. Zhang, *AIP Adv.* **2017**, 7, 075210.
- [121] L. Wang, F. Gao, K. Zhang, M. Wang, M. Qin, J. Kong, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24, 704.
- [122] M. Arbatti, X. Shan, Z. Cheng, *Adv. Mater.* **2007**, 19, 1369.
- [123] H. Tang, Z. Zhou, C. C. Bowland, H. A. Sodano, *Nano Energy* **2015**, 17, 302.
- [124] H. Tang, M. H. Malakooti, H. A. Sodano, *Appl. Phys. Lett.* **2013**, 103, 222901.
- [125] H. Tang, Y. Lin, C. Andrews, H. A. Sodano, *Nanotechnology* **2011**, 22, 15702.
- [126] H. Tang, Y. Lin, H. A. Sodano, *Adv. Energy Mater.* **2012**, 2, 469.
- [127] Y. Sun, Z. Zhang, C. P. Wong, *Polymer* **2005**, 46, 2297.
- [128] L. Y. Jiang, C. M. Leu, K. H. Wei, *Adv. Mater.* **2002**, 14, 426.
- [129] T. Tanaka, Y. Ohki, M. Ochi, M. Harada, T. Imai, *IEEE Trans. Dielectr. Electr. Insul.* **2008**, 15, 81.
- [130] V. Tomer, G. Polizos, C. A. Randall, E. Manias, *J. Appl. Phys.* **2011**, 109, 074113.
- [131] V. Tomer, E. Manias, C. A. Randall, *J. Appl. Phys.* **2011**, 110, 044107.
- [132] G. Polizos, V. Tomer, E. Manias, C. A. Randall, *J. Appl. Phys.* **2010**, 108, 074117.
- [133] V. Tomer, G. Polizos, E. Manias, C. A. Randall, *J. Appl. Phys.* **2010**, 108, 074116.
- [134] S. P. Fillery, H. Koerner, L. Drummy, E. Dunkerley, M. F. Durstock, D. F. Schmidt, R. A. Vaia, *ACS Appl. Mater. Interfaces* **2012**, 4, 1388.
- [135] Q. Li, G. Zhang, F. Liu, K. Han, M. R. Gadinski, C. Xiong, Q. Wang, *Energy Environ. Sci.* **2015**, 8, 922.
- [136] Q. Li, K. Han, M. R. Gadinski, G. Zhang, Q. Wang, *Adv. Mater.* **2014**, 26, 6244.
- [137] X. Huang, C. Zhi, P. Jiang, D. Golberg, Y. Bando, T. Tanaka, *Adv. Funct. Mater.* **2013**, 23, 1824.
- [138] Z. Wang, T. Iizuka, M. Kozako, Y. Ohki, T. Tanaka, *IEEE Trans. Dielectr. Electr. Insul.* **2011**, 18, 1963.
- [139] F. Liu, Q. Li, J. Cui, Z. Li, G. Yang, Y. Liu, L. Dong, C. Xiong, H. Wang, Q. Wang, *Adv. Funct. Mater.* **2017**, 27, 1606292.
- [140] A. Azizi, M. R. Gadinski, Q. Li, M. Abu AlSaud, J. Wang, Y. Wang, B. Wang, F. Liu, L. Chen, N. Alem, Q. Wang, *Adv. Mater.* **2017**, 29, 1701864.
- [141] Y. Zhan, Z. Long, X. Wan, C. Zhan, J. Zhang, Y. He, *Ceram. Int.* **2017**, 43, 12109.
- [142] F. Liu, Q. Li, Z. Li, Y. Liu, L. Dong, C. Xiong, Q. Wang, *Compos. Sci. Technol.* **2017**, 142, 139.
- [143] J. Yang, H. Xie, H. Chen, Z. Shi, T. Wu, Q. Yang, C. Xiong, *J. Mater. Chem. A* **2018**, 6, 1403.
- [144] G. Wang, X. Huang, P. Jiang, *ACS Appl. Mater. Interfaces* **2015**, 7, 18017.
- [145] Z. Shen, J. Wang, Y. Lin, C. W. Nan, L. Chen, Y. Shen, *Adv. Mater.* **2018**, 30, 1704380.
- [146] Y. Song, Y. Shen, H. Y. Liu, Y. H. Lin, M. Li, C. W. Nan, *J. Mater. Chem.* **2012**, 22, 8063.
- [147] B. Xie, H. Zhang, Q. Zhang, J. Zang, C. Yang, Q. Wang, M. Li, S. Jiang, *J. Mater. Chem. A* **2017**, 5, 6070.
- [148] X. Zhang, W. Chen, J. Wang, Y. Shen, L. Gu, Y. Lin, C. W. Nan, *Nanoscale* **2014**, 6, 6701.
- [149] G. Zhang, X. Zhang, H. Huang, J. Wang, Q. Li, L. Chen, Q. Wang, *Adv. Mater.* **2016**, 28, 4811.
- [150] Z. Pan, B. Liu, J. Zhai, L. Yao, K. Yang, B. Shen, *Nano Energy* **2017**, 40, 587.
- [151] S. Kango, S. Kalia, A. Celli, J. Njuguna, Y. Habibi, R. Kumar, *Prog. Polym. Sci.* **2013**, 38, 1232.
- [152] M. Iijima, H. Kamiya, *KONA Powder Part. J.* **2009**, 27, 119.
- [153] B. Hallouet, P. Desclaux, B. Wetzel, A. K. Schlarb, R. Pelster, *J. Phys. D: Appl. Phys.* **2009**, 42, 064004.
- [154] T. Andritsch, R. Kochetov, P. H. F. Morshuis, J. J. Smit, *Annu. Rep. - Conf. Electr. Insul. Dielectr. Phenom.* **2011**, 624.
- [155] P. Martins, C. M. Costa, M. Benelmekki, G. Botelho, S. Lanceros-Mendez, *CrystEngComm* **2012**, 14, 2807.
- [156] M. Roy, J. K. Nelson, C. W. Reed, R. K. MacCrone, R. J. Keefe, W. Zenger, L. S. Schadler, *IEEE Trans. Dielectr. Electr. Insul.* **2005**, 12, 1273.

- [157] S. Li, G. Yin, S. Bai, J. Li, *IEEE Trans. Dielectr. Electr. Insul.* **2011**, 18, 1535.
- [158] E. Kubyskhina, B. L. G. Jonsson, M. Unge, *IEEE Electr. Insul. Conf.* **2015**, 618.
- [159] R. C. Smith, L. Hui, J. K. Nelson, L. S. Schadler, *Annu. Rep. - Conf. Electr. Insul. Dielectr. Phenom.* **2009**, 650.
- [160] D. Fabiani, G. C. Montanari, C. Laurent, G. Teyssedre, P. H. F. Morshuis, R. Bodega, L. A. Dissado, A. Campus, U. H. Nilsson, *IEEE Electr. Insul. Mag.* **2007**, 23, 11.
- [161] T. Takada, Y. Hayase, Y. Tanaka, T. Okamoto, *IEEE Trans. Dielectr. Electr. Insul.* **2008**, 15, 152.
- [162] S. Li, D. Min, W. Wang, G. Chen, *IEEE Trans. Dielectr. Electr. Insul.* **2016**, 23, 2777.
- [163] S. Ding, S. Yu, X. Zhu, S. Xie, R. Sun, W. Liao, C. Wong, *Appl. Phys. Lett.* **2017**, 111, 153902.
- [164] Y. Thakur, M. H. Lean, Q. M. Zhang, *Appl. Phys. Lett.* **2017**, 110, 122905.
- [165] M. Roy, C. W. Reed, R. K. MacCrone, L. S. Schadler, J. K. Nelson, R. Keefe, W. Zenger, *Proc. Int. Symp. Electr. Insul. Mater.* **2005**, 1, 223.
- [166] M. Roy, J. K. Nelson, R. K. MacCrone, L. S. Schadler, *J. Mater. Sci.* **2007**, 42, 3789.
- [167] B. Venkatesulu, B. L. G. Jonsson, H. Edin, M. Norgren, *IEEE Trans. Dielectr. Electr. Insul.* **2013**, 20, 177.
- [168] P. Murugaraj, D. Mainwaring, N. Mora-Huertas, *J. Appl. Phys.* **2005**, 98, 054304.
- [169] L. Yu, V. Ranjan, M. B. Nardelli, J. Bernholc, *Phys. Rev. B* **2009**, 80, 165432.
- [170] Z. Shen, J. Wang, X. Zhang, Y. Lin, C. W. Nan, L. Chen, Y. Shen, *Appl. Phys. Lett.* **2017**, 111, 92901.
- [171] E. Kubyskhina, M. Unge, B. L. G. Jonsson, *J. Chem. Phys.* **2017**, 146, 51101.
- [172] R. Borgani, L. K. H. Paon, M. S. Hedenqvist, U. W. Gedde, D. B. Haviland, *Nano Lett.* **2016**, 16, 5934.
- [173] N. Shi, R. Ramprasad, *IEEE Trans. Dielectr. Electr. Insul.* **2008**, 15, 170.
- [174] S. Peng, Q. Zeng, X. Yang, J. Hu, X. Qiu, J. He, *Sci. Rep.* **2016**, 6, 38978.
- [175] E. J. W. Verwey, *J. Phys. Colloid Chem.* **1947**, 51, 631.
- [176] T. Missana, A. Adell, *J. Colloid Interface Sci.* **2000**, 230, 150.
- [177] M. Bostrom, V. Deniz, G. V. Franks, B. W. Ninham, *Adv. Colloid Interface Sci.* **2006**, 123, 5.
- [178] G. Chen, X. Wang, J. Lin, W. Yang, H. Li, Y. Wen, *J. Phys. Chem. C* **2016**, 120, 28423.
- [179] T. Zhou, J. Zha, R. Cui, B. Fan, J. Yuan, Z. Dang, *ACS Appl. Mater. Interfaces* **2011**, 3, 2184.
- [180] S. Chang, W. Liao, C. Ciou, J. Lee, C. Li, *J. Colloid Interface Sci.* **2009**, 329, 300.
- [181] M. N. Almadhoun, U. S. Bhansali, H. N. Alshareef, *J. Mater. Chem.* **2012**, 22, 11196.
- [182] H. Luo, D. Zhang, C. Jiang, X. Yuan, C. Chen, K. Zhou, *ACS Appl. Mater. Interfaces* **2015**, 7, 8061.
- [183] J. Li, S. I. Seok, B. Chu, F. Dogan, Q. Zhang, Q. Wang, *Adv. Mater.* **2009**, 21, 217.
- [184] Z. Wang, T. Wang, C. Wang, Y. Xiao, P. Jing, Y. Cui, Y. Pu, *ACS Appl. Mater. Interfaces* **2017**, 9, 29130.
- [185] Y. Xie, Y. Yu, Y. Feng, W. Jiang, Z. Zhang, *ACS Appl. Mater. Interfaces* **2017**, 9, 2995.
- [186] M. Iijima, N. Sato, I. Wuled Lenggoro, H. Kamiya, *Colloids Surf., A* **2009**, 352, 88.
- [187] Z. Dang, H. Wang, H. Xu, *Appl. Phys. Lett.* **2006**, 89, 112902.
- [188] C. Pan, K. Kou, Q. Jia, Y. Zhang, G. Wu, T. Ji, *Composites, Part B* **2017**, 111, 83.
- [189] F. Chao, G. Liang, *J. Mater. Sci.: Mater. Electron.* **2009**, 20, 560.
- [190] M. Iijima, M. Tsukada, H. Kamiya, *J. Colloid Interface Sci.* **2007**, 305, 315.
- [191] T. S. Sasikala, S. Raman, P. Mohanan, C. Pavithran, M. T. Sebastian, *J. Polym. Res.* **2011**, 18, 811.
- [192] L. Ramajo, M. S. Castro, M. M. Reboredo, *Composites, Part A* **2007**, 38, 1852.
- [193] S. Thomas, S. Raman, P. Mohanan, M. T. Sebastian, *Composites, Part A* **2010**, 41, 1148.
- [194] L. Xie, X. Huang, K. Yang, S. Li, P. Jiang, *J. Mater. Chem. A* **2014**, 2, 5244.
- [195] K. Yang, X. Huang, M. Zhu, L. Xie, T. Tanaka, P. Jiang, *ACS Appl. Mater. Interfaces* **2014**, 6, 1812.
- [196] L. A. Fredin, Z. Li, M. T. Lanagan, M. A. Ratner, T. J. Marks, *ACS Nano* **2013**, 7, 396.
- [197] Z. Li, L. A. Fredin, P. Tewari, S. A. DiBenedetto, M. T. Lanagan, M. A. Ratner, T. J. Marks, *Chem. Mater.* **2010**, 22, 5154.
- [198] L. A. Fredin, Z. Li, M. A. Ratner, M. T. Lanagan, T. J. Marks, *Adv. Mater.* **2012**, 24, 5946.
- [199] L. A. Fredin, Z. Li, M. T. Lanagan, M. A. Ratner, T. J. Marks, *Adv. Funct. Mater.* **2013**, 23, 3560.
- [200] L. Xie, X. Huang, C. Wu, P. Jiang, *J. Mater. Chem.* **2011**, 21, 5897.
- [201] S. A. Paniagua, Y. Kim, K. Henry, R. Kumar, J. W. Perry, S. R. Marder, *ACS Appl. Mater. Interfaces* **2014**, 6, 3477.
- [202] K. Yang, X. Huang, Y. Huang, L. Xie, P. Jiang, *Chem. Mater.* **2013**, 25, 2327.
- [203] D. Liu, A. T. Hoang, A. M. Pourrahimi, L. K. H. Pallon, F. Nilsson, S. M. Gubanski, T. Olsson, M. S. Hedenqvist, U. W. Gedde, *IEEE Trans. Dielectr. Electr. Insul.* **2017**, 24, 1396.
- [204] Y. Niu, Y. Bai, K. Yu, Y. Wang, F. Xiang, H. Wang, *ACS Appl. Mater. Interfaces* **2015**, 7, 24168.
- [205] S. Siddabattuni, T. P. Schuman, F. Dogan, *Mater. Sci. Eng., B* **2011**, 176, 1422.
- [206] A. M. Pourrahimi, T. A. Hoang, D. Liu, L. K. H. Pallon, S. Gubanski, R. T. Olsson, U. W. Gedde, M. S. Hedenqvist, *Adv. Mater.* **2016**, 28, 8651.
- [207] Y. Li, T. M. Krentz, L. Wang, B. C. Benicewicz, L. S. Schadler, *ACS Appl. Mater. Interfaces* **2014**, 6, 6005.
- [208] M. Bell, T. Krentz, J. K. Nelson, L. Schadler, K. Wu, C. Breneman, S. Zhao, H. Hillborg, B. Benicewicz, *J. Colloid Interface Sci.* **2017**, 495, 130.
- [209] S. Siddabattuni, T. P. Schuman, F. Dogan, *ACS Appl. Mater. Interfaces* **2013**, 5, 1917.
- [210] R. M. Diebold, M. J. Gordon, D. R. Clarke, *ACS Appl. Mater. Interfaces* **2014**, 6, 11932.
- [211] L. Li, R. Feng, Y. Zhang, L. Dong, *J. Mater. Chem. C* **2017**, 5, 11403.
- [212] R. C. Smith, L. Hui, J. K. Nelson, L. S. Schadler, *Annu. Rep. - Conf. Electr. Insul. Dielectr. Phenom.* **2009**, 129.
- [213] V. A. Markel, *J. Opt. Soc. Am. A* **2016**, 33, 1244.
- [214] M. Y. Koledintseva, R. E. DuBroff, R. W. Schwartz, *Prog. Electromagn. Res.* **2009**, 99, 131.
- [215] C. S. Daily, W. Sun, M. R. Kessler, X. Tan, N. Bowler, *IEEE Trans. Dielectr. Electr. Insul.* **2014**, 21, 488.
- [216] H. T. Vo, F. G. Shi, *Microelectron. J.* **2002**, 33, 409.
- [217] C. S. Daily, M. R. Kessler, X. Tan, N. Bowler, *Annu. Rep. - Conf. Electr. Insul. Dielectr. Phenom.* **2012**, 507.
- [218] D. Pitsa, M. G. Danikas, *IEEE Int. Conf. Solid Dielectr.* **2013**, 832.
- [219] Z. Su, X. Li, Y. Yin, *Proc. Int. Symp. Electr. Insul. Mater.* **2011**, 297.
- [220] M. G. Todd, F. G. Shi, *IEEE Trans. Dielectr. Electr. Insul.* **2005**, 12, 601.
- [221] Y. Rao, J. M. Qu, T. Marinis, C. P. Wong, *IEEE Trans. Compon. Packag. Technol.* **2000**, 23, 680.
- [222] Z. Dang, J. Yuan, J. Zha, T. Zhou, S. Li, G. Hu, *Prog. Mater. Sci.* **2012**, 57, 660.
- [223] G. Zhang, D. Brannum, D. Dong, L. Tang, E. Allahyarov, S. Tang, K. Kodweis, J. Lee, L. Zhu, *Chem. Mater.* **2016**, 28, 4646.
- [224] X. Zhang, Y. Shen, B. Xu, Q. Zhang, L. Gu, J. Jiang, J. Ma, Y. Lin, C. W. Nan, *Adv. Mater.* **2016**, 28, 2055.

- [225] Z. Pan, L. Yao, J. Zhai, D. Fu, B. Shen, H. Wang, *ACS Appl. Mater. Interfaces* **2017**, 9, 4024.
- [226] X. Lin, P. Hu, Z. Jia, S. Gao, *J. Mater. Chem. A* **2016**, 4, 2314.
- [227] S. Singha, M. J. Thomas, *IEEE Trans. Dielectr. Electr. Insul.* **2008**, 15, 12.
- [228] Y. Thakur, T. Zhang, C. Iacob, T. Yang, J. Bernholc, L. Q. Chen, J. Runt, Q. M. Zhang, *Nanoscale* **2017**, 9, 10992.
- [229] S. Javadi, M. Razzaghi-Kashani, P. N. B. Reis, A. A. Balado, *Polym. Compos.* **2017**, 38, 1158.
- [230] Y. U. Wang, D. Q. Tan, J. Krahn, *J. Appl. Phys.* **2011**, 110, 44103.
- [231] E. Tuncer, Y. V. Serdyuk, S. M. Gubanski, *IEEE Trans. Dielectr. Electr. Insul.* **2002**, 9, 809.
- [232] M. Yao, S. You, Y. Peng, *Ceram. Int.* **2017**, 43, 3127.
- [233] D. He, Y. Wang, X. Chen, Y. Deng, *Composites, Part A* **2017**, 93, 137.
- [234] D. He, Y. Wang, S. Song, S. Liu, Y. Luo, Y. Deng, *Compos. Sci. Technol.* **2017**, 151, 25.
- [235] Q. Huang, H. Luo, C. Chen, X. Zhou, K. Zhou, D. Zhang, *J. Alloys Compd.* **2017**, 696, 1220.
- [236] Q. Yuan, J. Cui, Y. Wang, R. Ma, H. Wang, *J. Eur. Ceram. Soc.* **2017**, 37, 4645.
- [237] M. Rahimabady, M. S. Mirshekarloo, K. Yao, L. Lu, *Phys. Chem. Chem. Phys.* **2013**, 15, 16242.
- [238] V. S. Puli, M. Ejaz, R. Elupula, M. Kothakonda, S. Adireddy, R. S. Katiyar, S. M. Grayson, D. B. Chrisey, *Polymer* **2016**, 105, 35.
- [239] J. Chen, X. Wang, X. Yu, L. Yao, Z. Duan, Y. Fan, Y. Jiang, Y. Zhou, Z. Pan, *J. Mater. Chem. C* **2018**, 6, 271.
- [240] D. He, Y. Wang, S. Song, S. Liu, Y. Deng, *ACS Appl. Mater. Interfaces* **2017**, 9, 44839.
- [241] J. Wang, Y. Long, Y. Sun, X. Zhang, H. Yang, B. Lin, *Appl. Surf. Sci.* **2017**, 426, 437.
- [242] W. Zhou, L. Xu, L. Jiang, J. Peng, Y. Gong, X. Liu, H. Cai, G. Wang, Q. Chen, *J. Alloys Compd.* **2017**, 710, 47.
- [243] L. Hu, Y. Xia, Q. Wang, H. Yang, Q. Zhang, *J. Mater. Sci.: Mater. Electron.* **2018**, 29, 1269.
- [244] K. Yu, Y. Niu, Y. Bai, Y. Zhou, H. Wang, *Appl. Phys. Lett.* **2013**, 102, 102903.
- [245] G. Wang, Y. Huang, Y. Wang, P. Jiang, X. Huang, *Phys. Chem. Chem. Phys.* **2017**, 19, 21058.
- [246] Z. Pan, J. Zhai, B. Shen, *J. Mater. Chem. A* **2017**, 5, 15217.